Lecture notes. Lecture 5

Unsupervised learning and phase diagrams of classical spin models

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1 Data without labels

1.1 Labels and prior knowledge

A magnet cooled from high temperature changes its behaviour at a critical temperature. Above it the spins point in random directions, below it most of them point the same way. For a new material a physicist wants to know how many such phases exist. A second question is where their boundaries lie in the plane of the control parameters, such as temperature and coupling constants. A further question is what kind of order sets in at each boundary. The raw material is a large set of measured or simulated snapshots of the microscopic state.

A neural network can be trained to classify such snapshots if each carries a label, for example "ordered" or "disordered". Labelling a snapshot as ordered presupposes that we know there are two phases, which observable distinguishes them and roughly where each phase lies. For a new model this knowledge is what we are trying to obtain. We examine how much of the structure of a phase diagram can be found from the snapshots alone, without labels.

1.2 A data set as a cloud of points

Without labels only the samples remain,

$$\mathcal{D} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}, \qquad \mathbf{x}_i \in \mathbb{R}^D, \tag{1.1}$$

where n is the number of samples and D the number of numbers per sample, the dimension of the data. A greyscale image of 28×28 pixels is a vector of $D = 784$ pixel values, one coordinate per pixel. A configuration of $N = L^2$ Ising spins $s_i = \pm 1$ on an $L \times L$ lattice is a vector in $\{-1, +1\}^N \subset \mathbb{R}^N$. We write x_{ij} for feature j, a pixel or a spin, of sample i, and stack the samples as the rows of the $n \times D$ data matrix X.

The samples are independent draws from an unknown distribution $p(\mathbf{x})$. Supervised learning estimates the conditional distribution $p(y \,|\, \mathbf{x})$ of a label y given a sample, and unsupervised learning asks about $p(\mathbf{x})$ itself. For a system in thermal equilibrium $p(\mathbf{x})$ is the Boltzmann distribution, known in principle. We want to learn its structure, the phases and the observables that distinguish them. Dimensionality reduction searches for a few combinations of the D numbers that describe most of the variation between samples. Clustering groups the samples, and anomaly detection flags samples that are unusual compared with a reference set. Each method proposes a structure, and we decide what it means.

1.3 The test bed of handwritten digits

Before the methods are applied to physics, each is tested on data whose structure is known, the MNIST collection (Modified National Institute of Standards and Technology) of handwritten digits [1]. It holds 60 000 training and 10 000 test images of 28×28 grey pixels, each pixel scaled to $[0, 1]$ and

each image labelled by its digit from 0 to 9. The labels are never given to the methods. They are used only afterwards, to colour points in a plot or to score whether a structure found without labels coincides with the ten digits. Maps and clusterings use a fixed random subset of 5000 test images.

The digits do not fill the space $\mathbb{R}^{784}$. An image of 784 independently random pixels looks like noise, never like a digit. Images of digits occupy a thin region in which neighbouring pixels are strongly correlated. Within it a few independent changes of a stroke, its thickness, slant or size, lead from one admissible image to another. Spin configurations at a given temperature are confined in a similar way, because the Boltzmann weight suppresses almost all of the 2^N configurations. Dimensionality reduction maps the data to a space of small dimension d,

$$\mathbf{x} \in \mathbb{R}^D \;\mapsto\; \mathbf{z} \in \mathbb{R}^d, \qquad d \ll D, \tag{1.2}$$

while keeping as much of their structure as possible, and the simplest such map is linear.

2 Principal component analysis

2.1 Variance along a direction

The simplest low-dimensional description of a cloud of points is a flat one. If the samples lie close to a plane of dimension k through their mean, each is described, up to a small error, by its k coordinates in that plane. Principal component analysis (PCA) was introduced by Pearson as the plane of closest fit [2] and by Hotelling as a few uncorrelated variables that carry the variation of many [3]. It chooses this plane in two equivalent ways, as the plane along which the data vary most and as the plane from which they deviate least.

All quantities are measured from the mean of the data, $\bar{\mathbf{x}} \coloneqq \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i$, for the digits the average image. How two features vary together is described by the $D \times D$ covariance matrix

$$C_{jl} \coloneqq \frac{1}{n}\sum_{i=1}^{n} (x_{ij} - \bar{x}_j)(x_{il} - \bar{x}_l), \qquad C = \frac{1}{n} X_c^{\top} X_c, \tag{2.1}$$

where X_c is the $n \times D$ matrix whose rows are the centred samples $(\mathbf{x}_i - \bar{\mathbf{x}})^{\top}$. The diagonal entry C_{jj} is the variance of feature j. An off-diagonal entry is positive when features j and l tend to be large in the same samples. It is negative when one tends to be large where the other is small. The matrix C is symmetric, and it is positive semidefinite, since $\mathbf{u}^{\top} C \mathbf{u} = \frac{1}{n}\|X_c \mathbf{u}\|^2 \geq 0$ for every vector $\mathbf{u}$.

Project every centred sample on a unit vector $\mathbf{u}$, $\mathbf{u}^{\top}\mathbf{u} = 1$, and call the result $w_i \coloneqq \mathbf{u}^{\top}(\mathbf{x}_i - \bar{\mathbf{x}})$. These numbers have mean zero, and their variance is

$$\frac{1}{n}\sum_{i=1}^{n} w_i^2 = \frac{1}{n}\sum_{i=1}^{n} \mathbf{u}^{\top}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^{\top}\mathbf{u} = \mathbf{u}^{\top} C\, \mathbf{u}. \tag{2.2}$$

The covariance matrix therefore gives the variance of the data along every direction. The direction of largest variance maximises $\mathbf{u}^{\top} C \mathbf{u}$ under the constraint $\mathbf{u}^{\top}\mathbf{u} = 1$. With a Lagrange multiplier σ^2 for the constraint, the condition for a stationary point is

$$\frac{\partial}{\partial \mathbf{u}}\Big[\mathbf{u}^{\top} C \mathbf{u} - \sigma^2(\mathbf{u}^{\top}\mathbf{u} - 1)\Big] = 2C\mathbf{u} - 2\sigma^2\mathbf{u} = 0 \quad \Longrightarrow \quad C\mathbf{u} = \sigma^2 \mathbf{u}. \tag{2.3}$$

A stationary direction is an eigenvector of C, and the variance along it, $\mathbf{u}^{\top} C \mathbf{u} = \sigma^2$, is its eigenvalue. The variance is largest along the eigenvector with the largest eigenvalue.

A real symmetric matrix has D real eigenvalues and an orthonormal basis of eigenvectors. We order them by decreasing eigenvalue,

$$C\boldsymbol{\lambda}_l = \sigma_l^2 \boldsymbol{\lambda}_l, \qquad \boldsymbol{\lambda}_l^{\top}\boldsymbol{\lambda}_m = \delta_{lm}, \qquad \sigma_1^2 \geq \sigma_2^2 \geq \cdots \geq \sigma_D^2 \geq 0, \tag{2.4}$$

with δ_{lm} the Kronecker delta. The unit vector $\boldsymbol{\lambda}_l$ is the l-th principal direction and σ_l^2 is the variance of the data along it. The second direction follows from the same argument restricted to vectors orthogonal to $\boldsymbol{\lambda}_1$. Such a vector can be written as $\mathbf{u} = \sum_{l\geq 2} c_l \boldsymbol{\lambda}_l$ with $\sum_{l\geq 2} c_l^2 = 1$, and its variance $\mathbf{u}^\top C \mathbf{u} = \sum_{l\geq 2} \sigma_l^2 c_l^2$ is at most σ_2^2, with equality for $\mathbf{u} = \boldsymbol{\lambda}_2$. The projection of a centred sample on the l-th direction,

$$w_i^{(l)} \coloneqq \boldsymbol{\lambda}_l^\top (\mathbf{x}_i - \bar{\mathbf{x}}), \qquad \mathbf{w}_i \coloneqq \left[w_i^{(1)}, \ldots, w_i^{(k)}\right]^\top = \Lambda_k^\top (\mathbf{x}_i - \bar{\mathbf{x}}) \in \mathbb{R}^k, \tag{2.5}$$

is its coordinate along the l-th direction. The leading directions are the principal components, PC 1, PC 2 and so on, and the $D \times k$ matrix $\Lambda_k \coloneqq (\boldsymbol{\lambda}_1, \ldots, \boldsymbol{\lambda}_k)$ collects the first k of them. Coordinates along different directions are uncorrelated, $\frac{1}{n}\sum_i w_i^{(l)} w_i^{(m)} = \boldsymbol{\lambda}_l^\top C \boldsymbol{\lambda}_m = \sigma_m^2 \delta_{lm}$. PCA replaces the D correlated features by D uncorrelated ones ordered by variance, and dimensionality reduction keeps the first k. Once $\bar{\mathbf{x}}$ and Λ_k are computed, the linear map (2.5) applies to new samples as well. A convention fixes the sign of each eigenvector, which is arbitrary because $-\boldsymbol{\lambda}_l$ is an eigenvector whenever $\boldsymbol{\lambda}_l$ is one.

Figure 1 shows the construction for 400 points drawn from a correlated normal distribution in the plane. The eigenvalues are $\sigma_1^2 = 3.56$ and $\sigma_2^2 = 0.25$, so the first direction carries 93.4% of the variance and the second 6.6%.

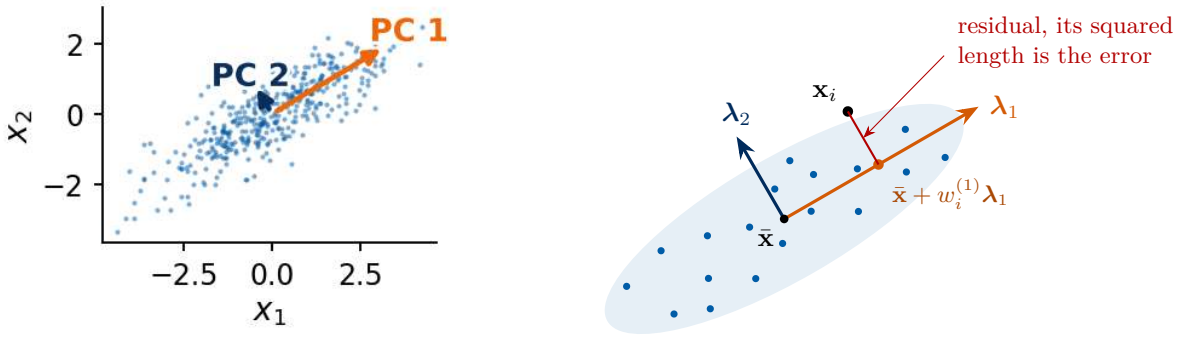


Figure 1: PCA of 400 points in the plane. Left, the centred points (coordinates x_1, x_2 after subtracting the mean) and the two principal directions. PC 1 runs along the long axis of the cloud and PC 2 across it, both drawn from the mean as arrows of length $2\sigma_1$ and $2\sigma_2$. Right, sketch. A sample $\mathbf{x}_i$ is projected on $\boldsymbol{\lambda}_1$ (orange point), which gives its coordinate $w_i^{(1)}$ and its reconstruction from one component. The perpendicular residual (red) is the part that the projection loses.

2.2 Minimal reconstruction error

The second view of PCA asks for the plane of dimension k through $\bar{\mathbf{x}}$ from which the data deviate least. Let the plane be spanned by the orthonormal columns of a $D \times k$ matrix R, $R^\top R = I_k$, where I_k is the identity matrix. The point of the plane closest to $\mathbf{x}_i$ is the orthogonal projection $\bar{\mathbf{x}} + RR^\top(\mathbf{x}_i - \bar{\mathbf{x}})$, and the residual $(I - RR^\top)(\mathbf{x}_i - \bar{\mathbf{x}})$ is perpendicular to the plane (Figure 1, right). Pythagoras' theorem splits the squared length of the centred sample,

$$\|\mathbf{x}_i - \bar{\mathbf{x}}\|^2 = \|R^\top(\mathbf{x}_i - \bar{\mathbf{x}})\|^2 + \|(I - RR^\top)(\mathbf{x}_i - \bar{\mathbf{x}})\|^2. \tag{2.6}$$

Averaging over the samples and using equation (2.2) for each column of R, the mean squared reconstruction error is

$$\mathcal{E}(R) \coloneqq \frac{1}{n}\sum_{i=1}^{n} \|(I - RR^\top)(\mathbf{x}_i - \bar{\mathbf{x}})\|^2 = \operatorname{tr} C - \operatorname{tr}\left(R^\top C R\right), \tag{2.7}$$

where $\operatorname{tr} C = \sum_j C_{jj}$ is the total variance, independent of the plane. Minimising the reconstruction error is the same as maximising $\operatorname{tr}(R^\top C R)$, the variance captured in the plane, so the two views

of PCA are one optimisation problem. To find the maximum, expand each column $\mathbf{r}_a$ of R in the eigenbasis of C. Then

$$\operatorname{tr}\left(R^\top C R\right) = \sum_{a=1}^{k} \mathbf{r}_a^\top C \mathbf{r}_a = \sum_{l=1}^{D} \sigma_l^2 \, c_l, \qquad c_l \coloneqq \sum_{a=1}^{k} \left(\boldsymbol{\lambda}_l^\top \mathbf{r}_a\right)^2 = \|R^\top \boldsymbol{\lambda}_l\|^2. \tag{2.8}$$

Each weight c_l is the squared length of the projection of the unit vector $\boldsymbol{\lambda}_l$ on the plane, so $0 \le c_l \le 1$, and the weights add up to $\sum_l c_l = \operatorname{tr}(R^\top R) = k$. A sum $\sum_l \sigma_l^2 c_l$ with decreasing σ_l^2 and such weights is largest when the weight 1 sits on the k largest eigenvalues. The maximum is reached for $R = \Lambda_k$, and the minimal reconstruction error is the sum of the discarded eigenvalues,

$$\min_R \mathcal{E}(R) = \mathcal{E}(\Lambda_k) = \sum_{l=k+1}^{D} \sigma_l^2. \tag{2.9}$$

The reconstruction of a sample from its first k coordinates is $\tilde{\mathbf{x}}_i = \bar{\mathbf{x}} + \Lambda_k \mathbf{w}_i = \bar{\mathbf{x}} + \sum_{l \le k} w_i^{(l)} \boldsymbol{\lambda}_l$, and the fraction of the variance that it keeps is

$$r_k \coloneqq \frac{\sum_{l \le k} \sigma_l^2}{\sum_{l \le D} \sigma_l^2} = 1 - \frac{\mathcal{E}(\Lambda_k)}{\operatorname{tr} C}. \tag{2.10}$$

A plot of $\sigma_l^2 / \sum_m \sigma_m^2$ against l, the spectrum of the data, shows how fast the variance is exhausted. A few dominant eigenvalues indicate a nearly flat structure of low dimension, and a flat spectrum indicates that no plane of small dimension describes the data.

2.3 The singular value decomposition

In practice the eigenvectors are computed from the singular value decomposition (SVD) of the centred data matrix, without forming C. Every real $n \times D$ matrix can be written as

$$X_c = U S V^\top = \sum_{l=1}^{m} s_l \, \mathbf{e}_l \mathbf{f}_l^\top, \qquad m = \min(n, D), \tag{2.11}$$

where $U = (\mathbf{e}_1, \ldots, \mathbf{e}_m)$ and $V = (\mathbf{f}_1, \ldots, \mathbf{f}_m)$ have orthonormal columns and $S = \operatorname{diag}(s_1, \ldots, s_m)$ holds the singular values $s_1 \ge s_2 \ge \cdots \ge 0$. Inserting the decomposition into equation (2.1) gives $C = V \, (S^2/n) \, V^\top$. The right singular vectors are the principal directions, $\mathbf{f}_l = \boldsymbol{\lambda}_l$, and the variances are $\sigma_l^2 = s_l^2 / n$. Forming C squares the ratio s_1/s_m of the largest to the smallest singular value, which measures how strongly rounding errors are amplified, so the SVD is the numerically safer route. Some implementations divide by $n - 1$ where we divide by n, which rescales all eigenvalues by one factor and leaves the directions and the fractions r_k unchanged.

The truncated decomposition $\sum_{l \le k} s_l \mathbf{e}_l \mathbf{f}_l^\top$ has the rows $(\tilde{\mathbf{x}}_i - \bar{\mathbf{x}})^\top$, the PCA reconstructions, and it is optimal among all matrices of rank k, a stronger statement than optimality among projections. Let F be any real $n \times D$ matrix with singular values $\varsigma_1 \ge \varsigma_2 \ge \ldots$, and let F_k be its decomposition truncated to the first k terms. The Eckart–Young theorem [4] states that for every matrix Y of the same shape and of rank at most k

$$\|F - Y\|_F^2 \;\ge\; \|F - F_k\|_F^2 = \sum_{l > k} \varsigma_l^2, \tag{2.12}$$

where $\|A\|_F^2 \coloneqq \sum_{ij} A_{ij}^2$ is the squared Frobenius norm. The theorem also states that F_k is the unique minimiser when $\varsigma_k > \varsigma_{k+1}$. The proof reuses the derivation above. The rows of Y lie in some subspace of dimension at most k with an orthonormal basis R, and the closest such rows are the projections $FRR^\top$. The steps from equation (2.7) to equation (2.9), applied to the symmetric matrix $F^\top F$ with eigenvalues ς_l^2, then give $\|F - FRR^\top\|_F^2 \ge \sum_{l>k} \varsigma_l^2$. For $F = X_c$ the theorem says that no compression of the centred data into k numbers per sample by a linear map, followed by a linear reconstruction, can beat PCA.

2.4 PCA of handwritten digits

On the 60 000 training images the covariance matrix has 784×784 entries, and each principal direction is itself a 28×28 image (Figure 2). The spectrum falls quickly over the first components and slowly afterwards. Two components keep $r_2 = 16.8\%$ of the variance, three 23.0%, ten 48.8% and fifty 82.5%, and 90% needs 87 components, so the digits are far from a plane of small dimension. Equation (2.9) holds on these data. The discarded eigenvalues per pixel, $\frac{1}{784}\sum_{l>k}\sigma_l^2$, are 0.06072, 0.01179 and 0.00672 for $k = 1$, 50 and 87. The mean squared reconstruction errors per pixel of the 5000 test images, which were not used to compute the directions, are 0.06078, 0.01157 and 0.00661.

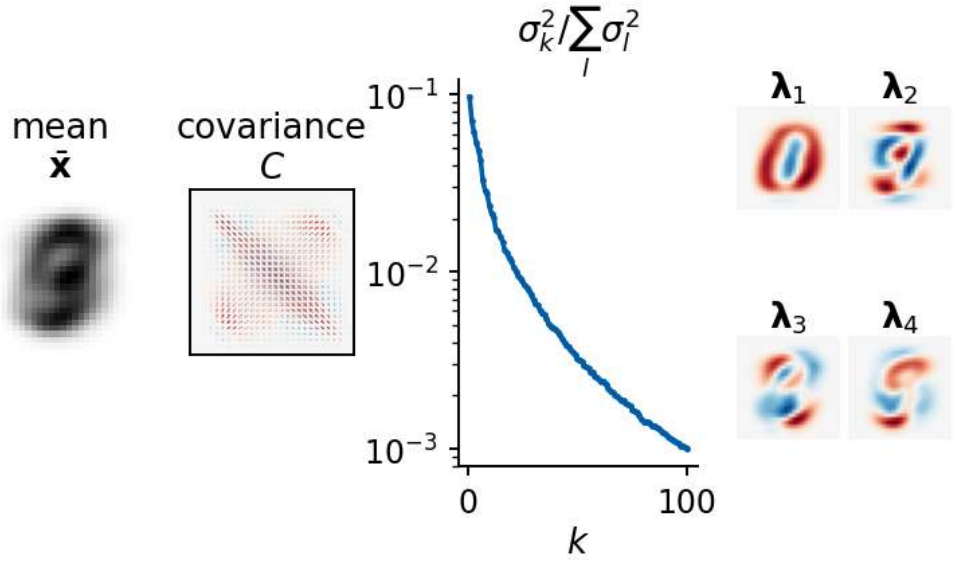


Figure 2: PCA of the 60 000 MNIST training images. From left to right, the mean image $\bar{\mathbf{x}}$, the covariance matrix C as a 784×784 image. Next come the fraction of the variance $\sigma_k^2 / \sum_l \sigma_l^2$ carried by the k-th component against k (logarithmic vertical axis) and the first four principal directions $\boldsymbol{\lambda}_1, \ldots, \boldsymbol{\lambda}_4$ as 28×28 images. Red marks positive and blue negative entries.

The coordinate $w^{(1)} = \sum_j \lambda_{1j}(x_j - \bar{x}_j)$ is a weighted sum of pixel deviations from the mean image, large when ink appears where $\boldsymbol{\lambda}_1$ is positive and absent where it is negative. The first direction in Figure 2 is a ring of one sign with the opposite sign in the centre. Its coordinate therefore separates digits with ink on a ring, such as a 0, from digits with ink only along a central stroke, such as a 1. In the plane of the first two coordinates (Figure 3) the 5000 test digits form one cloud. Coloured by the true digit, the 1s and the 0s sit at opposite ends of the first axis, while the other digits overlap. A third component separates a few more digits, and most still overlap.

2.5 Limits of a linear projection

PCA assumes that the structure of interest is a flat subspace. When the data lie near a curved set, a projection on a plane superposes points that are far apart along the set. When points on a closed curve are projected on the first principal direction (Figure 4), points from opposite sides of the curve receive the same coordinate. One number, the position along the curve, would describe every point. A human reader tells the digits apart at a glance, yet the best plane through them superposes most of them. This behaviour is expected if the digits lie on curved, possibly disconnected sets in 784 dimensions. We need a method that gives up the global linear map and keeps only the neighbourhood of every point.

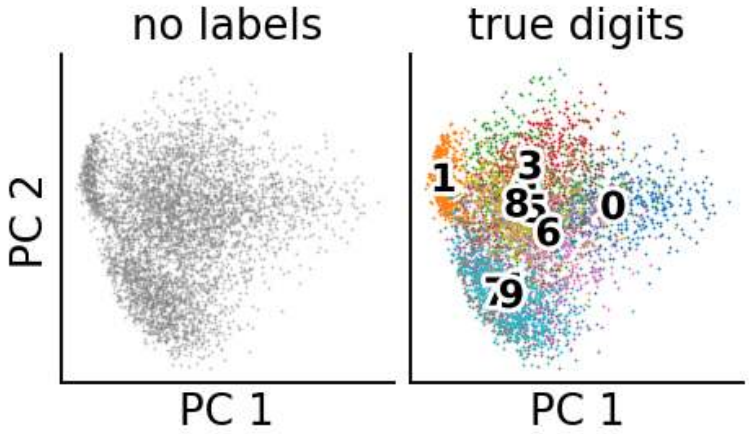


Figure 3: The 5000 MNIST test digits projected on the first two principal directions of the training images, coordinates $w^{(1)}$ (PC 1) and $w^{(2)}$ (PC 2). Left, the projection as PCA produces it, without labels. Right, the same points coloured by the true digit, with each digit written at the median of its points.

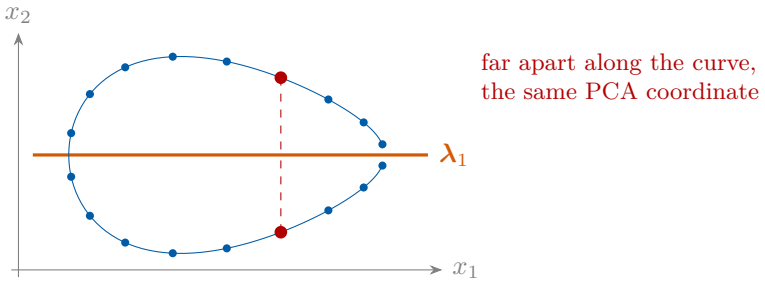


Figure 4: A limitation of PCA, sketch. Points on a closed curve (blue) are projected on the first principal direction $\boldsymbol{\lambda}_1$ (orange). Two points from opposite sides of the curve (red) receive nearly the same coordinate.

3 Maps that keep neighbours

3.1 Three hidden groups

Any map that keeps neighbouring points next to each other describes the curve of Figure 4 correctly, even if it distorts large distances. The method of t-distributed stochastic neighbour embedding (t-SNE) [5] grew out of stochastic neighbour embedding (SNE) [6] and turns this idea into an optimisation problem. Every sample $\mathbf{x}_i$ is represented by a point $\mathbf{y}_i \in \mathbb{R}^d$ of a map, usually with $d = 2$ so that the map can be plotted. Neighbourhood relations are described by probabilities, once among the data and once among the map points, and the map points are moved until the two descriptions agree. The name refers to this probabilistic, or stochastic, definition of neighbours and to the Student-t distribution used in the map.

Figure 5 shows the result for 450 unlabelled points in $\mathbb{R}^3$. They form three groups, uniform balls of radius 1 with centres 2.4 apart on the diagonal of the coordinate system. Seen along the diagonal the groups hide behind each other, and their values overlap along every coordinate axis, so no coordinate reveals them. In three dimensions the eye could still find a better viewing angle, which in a hundred dimensions is impossible. t-SNE with $d = 1$ assigns each point one number y_i, and three separate peaks appear on this line. We cut the line at its two largest gaps and obtain three groups, each containing points of only one true group. The purity, the fraction of points in each found group that belong to its most frequent true group, is 1.000. The line records only which points are neighbours, so the gaps between the peaks say little about the distances between the groups.

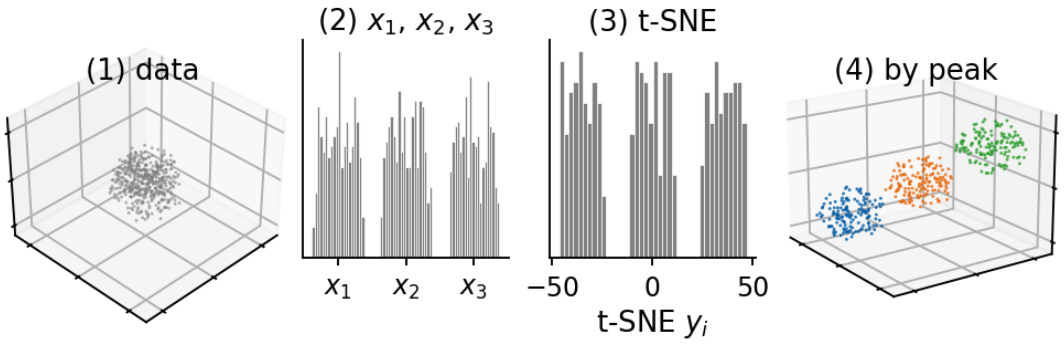


Figure 5: t-SNE of 450 points in $\mathbb{R}^3$ given without labels. (1) The cloud seen along the diagonal $(1,1,1)$. (2) Histograms of the three coordinates x_1, x_2, x_3, drawn side by side. (3) Histogram of the single number y_i that t-SNE with $d = 1$ assigns to each point. (4) The points in three dimensions, coloured by the peak of panel (3) to which they belong.

3.2 Similarities in the data

Around each sample $\mathbf{x}_i$ we place a Gaussian of width σ_i. We then define the probability that $\mathbf{x}_i$ would choose $\mathbf{x}_j$ as its neighbour if neighbours were chosen in proportion to the Gaussian weight,

$$p_{j|i} := \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2/2\sigma_i^2)}{\sum_{l\neq i}\exp(-\|\mathbf{x}_i - \mathbf{x}_l\|^2/2\sigma_i^2)}, \qquad p_{i|i} := 0, \tag{3.1}$$

where the squared distance $\|\mathbf{x}_i - \mathbf{x}_j\|^2 = \sum_{k=1}^{D}(x_{ik} - x_{jk})^2$ is summed over the D features. For fixed i the numbers $p_{j|i}$ form a probability distribution P_i over the $n-1$ other samples. The width σ_i, indexed by a sample, is unrelated to the PCA variances σ_l^2.

A common width for all samples fails, since in a dense region a small width already includes many neighbours and in a sparse region it includes none. t-SNE therefore fixes the effective number of neighbours of every sample, the perplexity,

$$\mathrm{Perp}(P_i) := e^{H(P_i)}, \qquad H(P_i) := -\sum_{j\neq i} p_{j|i}\ln p_{j|i}, \tag{3.2}$$

where H is the Shannon entropy of P_i. If $\mathbf{x}_i$ had k equally likely neighbours and no others, then $p_{j|i} = 1/k$, $H = \ln k$ and the perplexity would be k, so the perplexity counts neighbours. The entropy grows monotonically with σ_i, and bisection finds, for each sample separately, the width that gives the prescribed perplexity. For the 5000 test digits, represented by their 50 leading principal components, perplexity 30 gives widths from 0.619 to 2.263 with the median 1.451. The 1s, each a single straight stroke, lie close to each other and receive smaller widths than the 0s. Since $\sigma_i \neq \sigma_j$ in general, the conditional probabilities are asymmetric, and t-SNE uses the symmetric joint probabilities

$$p_{ij} := \frac{p_{j|i} + p_{i|j}}{2n}, \qquad \sum_{i\neq j} p_{ij} = 1, \tag{3.3}$$

which form one probability distribution P over all pairs of samples. Each of its entries is the probability that i and j are neighbours in the original space.

3.3 Similarities in the map and the crowding problem

In the map the similarity of two points is measured by an analogous distribution Q over pairs. A Gaussian kernel in the map, the choice of the original SNE, fails for a geometric reason, the crowding problem. In D dimensions up to $D+1$ points can lie at equal mutual distances, the vertices of a regular simplex, while in the plane at most three can (Figure 6). When high-dimensional neighbourhoods are flattened into the plane, many samples at moderate distances from $\mathbf{x}_i$ must be placed much farther

from $\mathbf{y}_i$. A Gaussian assigns such distant pairs a similarity that is essentially zero, while their p_{ij} is small and finite. The mismatch pulls all groups together into the centre of the map. t-SNE therefore uses a kernel with a heavy tail, the Student-t distribution with one degree of freedom,

$$q_{ij} \coloneqq \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{l \neq m} (1 + \|\mathbf{y}_l - \mathbf{y}_m\|^2)^{-1}}, \qquad q_{ii} \coloneqq 0, \tag{3.4}$$

where $\|\mathbf{y}_i - \mathbf{y}_j\|^2$ is summed over the d map coordinates. The denominator is summed over all pairs, so that Q is a probability distribution over pairs like P. The map kernel has no width, since a width would only rescale the free map coordinates. At small distances the two kernels are similar (Figure 7), whereas at large distances the Gaussian decays as $e^{-r^2/2}$ and the Student-t only as $1/r^2$. A pair with a small but nonzero p_{ij} can therefore sit far apart in the map and still receive a matching q_{ij}, and gaps open between groups.

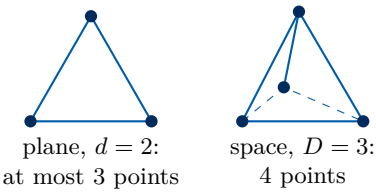


Figure 6: The crowding problem, sketch. Equal mutual distances are realised by $D + 1$ points in D dimensions but only by three in the plane.

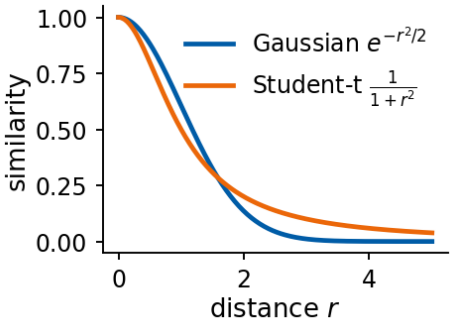


Figure 7: Similarity against the distance r for the Gaussian kernel $e^{-r^2/2}$ of the data and the Student-t kernel $1/(1 + r^2)$ of the map.

3.4 The loss and its gradient

The map is found by making Q as close as possible to P. The measure of closeness is the Kullback–Leibler (KL) divergence,

$$\mathrm{KL}(P \,\|\, Q) \coloneqq \sum_{i \neq j} p_{ij} \ln \frac{p_{ij}}{q_{ij}}, \tag{3.5}$$

a function of all map points $\mathbf{y}_1, \ldots, \mathbf{y}_n$ through the q_{ij}. It is nonnegative and vanishes only for $Q = P$, by Jensen's inequality for the concave logarithm, $-\mathrm{KL} = \sum p_{ij} \ln(q_{ij}/p_{ij}) \leq \ln \sum q_{ij} = 0$. The asymmetry of the divergence determines what the map preserves. A pair of neighbours placed far apart in the map has a large p_{ij} and a small q_{ij} and contributes a large term. A pair of distant samples placed close together has a small p_{ij}, and its term is small whatever q_{ij} is. The loss therefore keeps neighbours together and barely constrains the distances between non-neighbours, so t-SNE keeps the local structure and gives up the global one.

The map points are found by gradient descent. Write $\tilde{q}_{ij} \coloneqq (1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}$, so that $q_{ij} = \tilde{q}_{ij} / \sum_{l \neq m} \tilde{q}_{lm}$. Using $\sum_{i \neq j} p_{ij} = 1$, the loss is $\mathrm{KL} = \mathrm{const} - \sum_{i \neq j} p_{ij} \ln \tilde{q}_{ij} + \ln \sum_{l \neq m} \tilde{q}_{lm}$. The point $\mathbf{y}_i$ enters the pairs (i, j) and (j, i), and $\partial \tilde{q}_{ij} / \partial \mathbf{y}_i = -2\tilde{q}_{ij}^2 (\mathbf{y}_i - \mathbf{y}_j)$. The second term contributes $4 \sum_j p_{ij} \tilde{q}_{ij} (\mathbf{y}_i - \mathbf{y}_j)$ and the third $-4 \sum_j q_{ij} \tilde{q}_{ij} (\mathbf{y}_i - \mathbf{y}_j)$, which together give

$$\frac{\partial\, \mathrm{KL}(P \,\|\, Q)}{\partial \mathbf{y}_i} = 4 \sum_{j \neq i} (p_{ij} - q_{ij}) \frac{\mathbf{y}_i - \mathbf{y}_j}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2}. \tag{3.6}$$

The gradient is a sum of pairwise forces. A step of gradient descent moves $\mathbf{y}_i$ towards $\mathbf{y}_j$ when $p_{ij} > q_{ij}$ and away from it when $p_{ij} < q_{ij}$. Pairs closer in the data than in the map attract, and pairs closer in the map repel. The factor $(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}$ weakens the force between distant map

points, so that well separated groups barely act on each other. Automatic differentiation of the loss reproduces this gradient to 1.16×10^{-9} in absolute value on 300 points.

Plain gradient descent works poorly on this nonconvex loss. The standard optimiser [5] adds momentum, a fraction of the previous update added to the current one, 0.5 during the first 250 iterations and 0.8 afterwards. During the same 250 iterations all p_{ij} are multiplied by 12, a step called early exaggeration. Attractions then dominate, and groups form while the map is still compact and points can move past each other. Each coordinate also has its own adaptive step size. The map starts from the first d principal components of the data scaled to a small spread, so that repeated runs give the same result. The sums over all pairs make one iteration cost of order n^2 operations, and faster implementations approximate the repulsion of distant groups with a Barnes–Hut tree [7].

3.5 Maps of digits

On the 5000 test digits, represented by their 50 leading principal components, t-SNE with perplexity 30 gives the map of Figure 8, with the final loss KL = 1.335. Without colours about ten islands appear, and the colours identify them with the digits. The 4s and 9s, which differ by a small stroke, form adjacent islands that touch. The map uses only the pixel values, so each digit forms a connected set of neighbours in pixel space, although no plane separates these sets. A map with $d = 3$ has more room and reaches the lower loss 1.215. The perplexity sets the scale of the neighbourhoods (Figure 9). At perplexity 5 only the closest neighbours enter and the groups break into fragments, and at 100 the map is smoother.

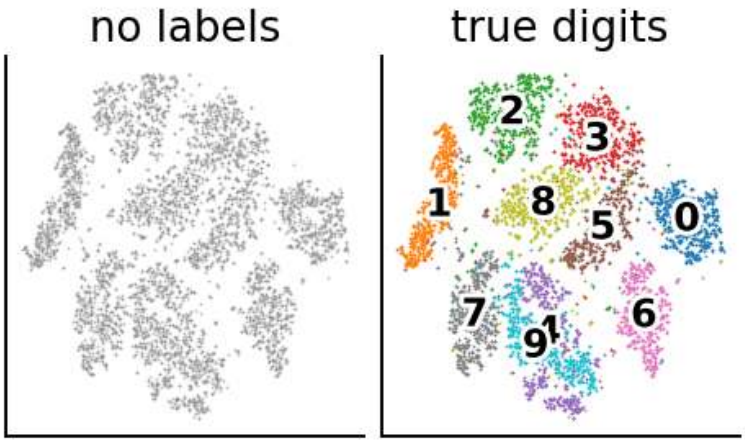


Figure 8: t-SNE map ($d = 2$, perplexity 30) of the 5000 MNIST test digits computed from their 50 leading principal components. The axes are the two map coordinates y_{i1} and y_{i2}, whose values carry no meaning. Left, the map without labels. Right, the same points coloured by the true digit, with each digit written at the median of its points.

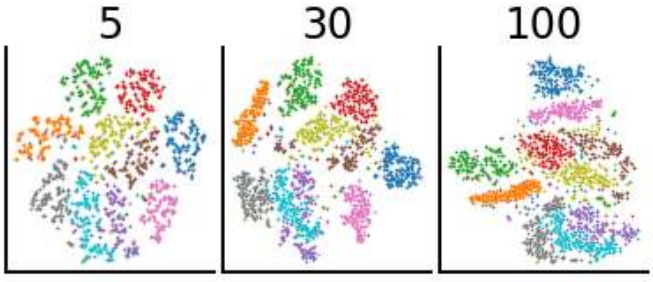


Figure 9: t-SNE maps of the first 2000 of the same digits at perplexity 5, 30 and 100, coloured by the true digit with the colours of Figure 8.

Two runs of the optimisation, or two different optimisers, give different maps, and a point-by-point comparison of their coordinates is meaningless. We compare them by what a t-SNE map is meant to preserve, the nearest neighbours. The fraction of the ten nearest map neighbours of a point that show the same digit, averaged over all points, uses the labels only for scoring. It is 0.916 for the exact optimisation and 0.918 for the Barnes–Hut approximation, whose map has the exact loss 1.379. In the 50 principal coordinates from which both maps were computed the same fraction is 0.896, so discarding the large distances makes the digit structure easier to see.

3.6 Content of a t-SNE map

The loss (3.5) determines which features of a map carry meaning [8]. It depends only on distances between map points, so a rotated, reflected or shifted map is equally good and the position of a group is arbitrary. The separation between groups says little, because pairs of distant samples have $p_{ij} \approx 0$ and hardly enter the loss while the heavy tail pushes groups apart. Group sizes say little as well, since the widths σ_i adapt to the local density. Inside a group the map keeps only the nearest neighbours, roughly as many as the perplexity, and distorts larger distances. Its reliable content is which samples are close neighbours (Figure 10). Since the map points are free parameters, a new sample has no map point until the map is recomputed, and no sample can be reconstructed from its map coordinates.

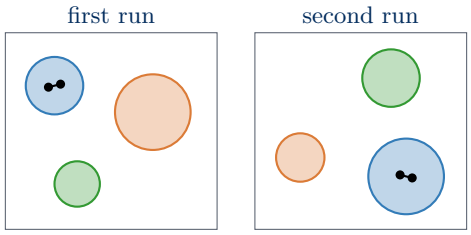


Figure 10: Two t-SNE runs on the same data with different starting maps, sketch. The groups (coloured discs) move, mirror and change size and separation, while two neighbouring samples (black) stay together.

PCA yields a linear map that applies to any new sample and from which the sample can be approximately rebuilt. t-SNE draws the neighbourhoods of the samples it was shown. Both end in plots inspected by eye. Counting the groups, assigning each sample to one of them and comparing them with a known classification require an explicit procedure that partitions the samples.

4 Groups of samples

4.1 k-means

A partition of the samples into k groups is good, in the simplest sense, if the samples of each group lie close to a common centre. The k-means method seeks disjoint sets of sample indices $S_1, \dots, S_k$, which together contain every sample once, and centres $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_k$. Sets and centres minimise the total squared distance of the samples to the centres of their groups,

$$\mathcal{L}_{\mathrm{km}} := \sum_{c=1}^{k} \sum_{i \in S_c} \|\mathbf{x}_i - \boldsymbol{\mu}_c\|^2. \tag{4.1}$$

The number of groups k is chosen in advance. The objective contains a discrete assignment of samples to groups and continuous centres, and Lloyd's algorithm [9] minimises it by alternating between them. In the assignment step every sample joins its nearest centre, $c_i = \arg\min_c \|\mathbf{x}_i - \boldsymbol{\mu}_c\|^2$, which minimises $\mathcal{L}_{\mathrm{km}}$ over the assignments for fixed centres. In the update step every centre moves to the mean of its samples,

$$\frac{\partial \mathcal{L}_{\mathrm{km}}}{\partial \boldsymbol{\mu}_c} = -2 \sum_{i \in S_c} (\mathbf{x}_i - \boldsymbol{\mu}_c) = 0 \quad \Longrightarrow \quad \boldsymbol{\mu}_c = \frac{1}{|S_c|} \sum_{i \in S_c} \mathbf{x}_i, \tag{4.2}$$

which minimises $\mathcal{L}_{\mathrm{km}}$ over the centres for fixed assignments and gives the method its name. Neither step increases $\mathcal{L}_{\mathrm{km}}$, and the number of partitions is finite, so the algorithm stops after finitely many iterations, when no sample changes its group. It ends in a local minimum that depends on the starting centres. The k-means++ initialisation [10] spreads the starting centres over the data. It takes a random sample as the first centre and draws every further centre among the samples with probability proportional to the squared distance from the nearest centre already chosen. We run the algorithm from 10 such starts and keep the partition with the smallest $\mathcal{L}_{\mathrm{km}}$. Because each sample joins its nearest centre, the groups are convex cells, and k-means favours round groups of similar extent.

Figure 11 follows the algorithm on three groups of 100 points in the plane, with all three starting centres placed inside one group. The objective falls from 9424 to 761.3, and iteration 4 changes no assignment. On less clearly separated data a bad start can persist to convergence.

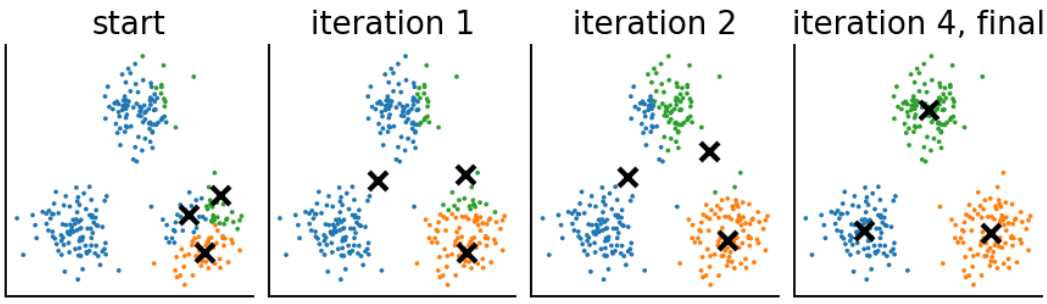


Figure 11: Lloyd's algorithm with $k = 3$ on 300 points in the plane. Colours show the current assignment of the points and crosses the current centres. The panels show the start (all three centres inside the lower right group), the state after the first two iterations and convergence after iteration 4.

4.2 Judging a partition

A clustering algorithm always returns groups, so a partition has to be judged. Without labels, the silhouette [11] compares the distance of a sample to its own group with the distance to the nearest other group. For a sample i in group S_c we write a_i for its mean distance to the other members of S_c. The number b_i is the smallest mean distance to the members of another group. The silhouette $h_i := (b_i - a_i)/\max(a_i, b_i) \in [-1, 1]$ is close to 1 for a sample well inside its group, close to 0 on a border and negative for a sample closer to another group. Its mean over all samples scores the partition. The objective (4.1) always decreases with k, whereas the mean silhouette has a maximum. For four Gaussian groups of 150 points the maximum, 0.692, lies at $k = 4$.

When labels exist, a partition can be compared with them. Group names are arbitrary, so the comparison counts pairs of samples. A pair agrees if both partitions put its two samples in the same group or both put them in different groups. The Rand index (RI) [12] is the fraction of agreeing pairs among all $\binom{n}{2}$. Everything follows from the contingency table n_{uv}, the number of samples in class u and group v, with row sums a_u and column sums b_v. The pairs together in both partitions number $\sum_{uv} \binom{n_{uv}}{2}$, and the pairs apart in both number $\binom{n}{2} - \sum_u \binom{a_u}{2} - \sum_v \binom{b_v}{2} + \sum_{uv} \binom{n_{uv}}{2}$. Random groups already agree on many pairs, because most pairs are apart in both partitions. The adjusted Rand index (ARI) [13] subtracts the mean $\mathbb{E}[\mathrm{RI}]$ over random relabellings that keep all group sizes,

$$\mathrm{ARI} := \frac{\mathrm{RI} - \mathbb{E}[\mathrm{RI}]}{1 - \mathbb{E}[\mathrm{RI}]} = \frac{\sum_{uv} \binom{n_{uv}}{2} - \sum_u \binom{a_u}{2} \sum_v \binom{b_v}{2} / \binom{n}{2}}{\frac{1}{2}\left[\sum_u \binom{a_u}{2} + \sum_v \binom{b_v}{2}\right] - \sum_u \binom{a_u}{2} \sum_v \binom{b_v}{2} / \binom{n}{2}}, \tag{4.3}$$

which is 1 for identical partitions, close to 0 for a random assignment, and can be negative. The correction is large (Figure 12), since random groups of four clouds reach RI = 0.626 and only ARI = 0.001.

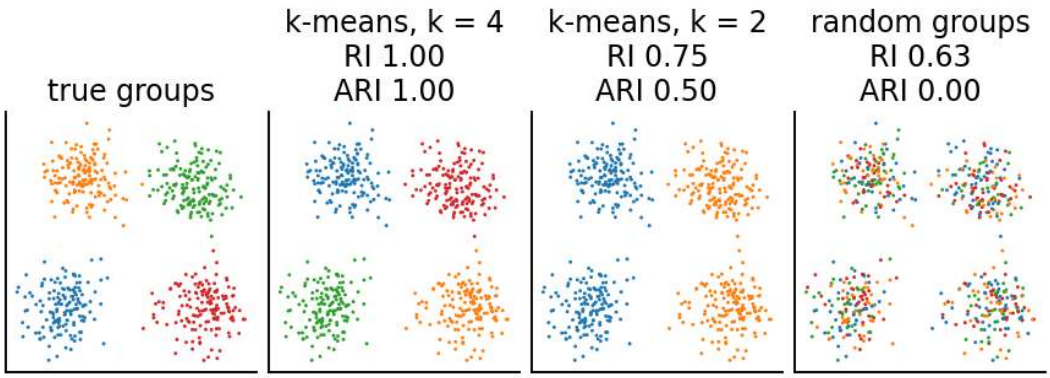


Figure 12: 600 points around four centres in the plane and three partitions of them, compared with the four true groups. From left to right, the panels show the true groups and k-means with $k = 4$ (RI = 0.998, ARI = 0.996, rounded to two decimals in the titles). The last two show k-means with $k = 2$ (RI = 0.748, ARI = 0.495) and random groups of the same sizes (RI = 0.626, ARI = 0.001). Colours mark the groups of each partition.

4.3 Groups of digits

Asked for $k = 10$ groups, k-means agrees with the digits with ARI = 0.380 on the 50 leading principal components. On the t-SNE map it reaches ARI = 0.757 (Figure 13), since the map provides the round groups of similar size that k-means looks for. In this run two groups contain mixtures of 4s and 9s. The silhouette on the t-SNE map is flat over a range of k around ten, with the maximum at $k = 9$ and the value 0.468 at $k = 10$. On real data the number of groups therefore remains a judgement of the analyst.

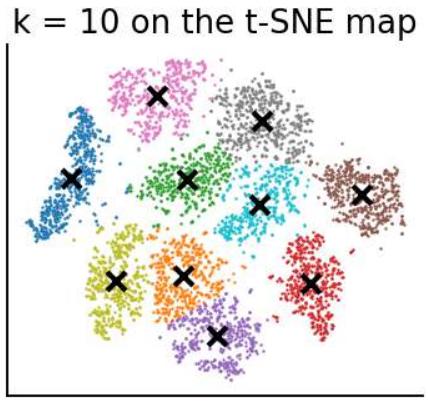


Figure 13: k-means with $k = 10$ on the t-SNE map of Figure 8. Colours mark the ten groups found without labels and crosses their centres. Compared with the true digits, ARI = 0.757.

A t-SNE map is a good input for k-means, although it describes only the samples it was given. A representation that applies to new samples, keeps more than a linear projection and rebuilds a sample from a few numbers requires a nonlinear map with an inverse.

5 Autoencoders

5.1 Encoder, decoder and bottleneck

PCA can be read as two linear maps fitted by minimising the reconstruction error (2.7). The first, $\mathbf{w} = \Lambda_k^\top(\mathbf{x} - \bar{\mathbf{x}})$, compresses a sample of D numbers into k numbers, and the second, $\tilde{\mathbf{x}} = \bar{\mathbf{x}} + \Lambda_k \mathbf{w}$, expands them back. The reconstructions of all samples lie on a flat k-dimensional plane, and Figure 4 showed what this costs when the data lie near a curved set. An autoencoder keeps the two maps and

replaces each by a neural network [14, 15], so that the set of possible reconstructions can be a curved surface that follows the data.

The encoder $E_{\boldsymbol{\theta}_E}\colon \mathbb{R}^D \to \mathbb{R}^d$, with parameters $\boldsymbol{\theta}_E$ (weights and biases), maps a sample to its latent vector $\mathbf{z} = E_{\boldsymbol{\theta}_E}(\mathbf{x})$. The decoder $G_{\boldsymbol{\theta}_G}\colon \mathbb{R}^d \to \mathbb{R}^D$, with parameters $\boldsymbol{\theta}_G$, maps the latent vector to the reconstruction $\tilde{\mathbf{x}} = G_{\boldsymbol{\theta}_G}(\mathbf{z})$. The space $\mathbb{R}^d$ of latent vectors is the latent space, and d plays the role of the number k of kept principal components. We measure the reconstruction error of one sample by

$$\varepsilon(\mathbf{x}) \coloneqq \frac{1}{D}\sum_{j=1}^{D}\left(x_j - \tilde{x}_j\right)^2, \qquad \tilde{\mathbf{x}} = G_{\boldsymbol{\theta}_G}\left(E_{\boldsymbol{\theta}_E}(\mathbf{x})\right), \tag{5.1}$$

the mean of the squared differences over its D pixels. The loss is its mean over the n training samples,

$$\mathcal{L}(\boldsymbol{\theta}_E, \boldsymbol{\theta}_G) \coloneqq \frac{1}{n}\sum_{i=1}^{n}\varepsilon(\mathbf{x}_i) = \frac{1}{nD}\sum_{i=1}^{n}\sum_{j=1}^{D}\left(x_{ij} - \tilde{x}_{ij}\right)^2, \tag{5.2}$$

the mean squared error (MSE) over all nD pixel values. Every input is its own target, so no label enters. Gradient descent updates $\boldsymbol{\theta}_E$ and $\boldsymbol{\theta}_G$ together, with the gradient flowing from the loss through the decoder into the encoder.

Without the bottleneck $d \ll D$ a network that copies its input would reach zero loss. With it, the d numbers of $\mathbf{z}$ must carry whatever the decoder needs to rebuild the sample, and the autoencoder performs a lossy compression with ratio D/d. Minimising the loss pushes the encoder to keep the features that vary most between samples and discard the rest, as PCA keeps the directions of largest variance. Because input and reconstruction are compared pixel by pixel, the loss rewards placing the bulk of the ink correctly, and features that occupy few pixels contribute little. It places no requirement on the latent space.

5.2 A dense autoencoder for the digits

The autoencoder trained on the digits has fully connected, or dense, layers. Its encoder narrows the 784 pixels in three layers to d latent numbers, and its decoder widens them back in three mirrored layers,

$$\mathbf{h}_1 = \mathrm{ReLU}\left(W_1\mathbf{x} + \mathbf{b}_1\right) \in \mathbb{R}^{256}, \tag{5.3a}$$
$$\mathbf{h}_2 = \mathrm{ReLU}\left(W_2\mathbf{h}_1 + \mathbf{b}_2\right) \in \mathbb{R}^{64}, \tag{5.3b}$$
$$\mathbf{z} = W_3\mathbf{h}_2 + \mathbf{b}_3 \in \mathbb{R}^{d}, \tag{5.3c}$$
$$\mathbf{h}_3 = \mathrm{ReLU}\left(W_4\mathbf{z} + \mathbf{b}_4\right) \in \mathbb{R}^{64}, \tag{5.3d}$$
$$\mathbf{h}_4 = \mathrm{ReLU}\left(W_5\mathbf{h}_3 + \mathbf{b}_5\right) \in \mathbb{R}^{256}, \tag{5.3e}$$
$$\tilde{\mathbf{x}} = \mathrm{sigm}\left(W_6\mathbf{h}_4 + \mathbf{b}_6\right) \in (0,1)^{784}, \tag{5.3f}$$

where the rectified linear unit $\mathrm{ReLU}(u) = \max(0, u)$ and the logistic function $\mathrm{sigm}(u) = 1/(1+e^{-u})$ act on each component. The encoder parameters are $\boldsymbol{\theta}_E = \{W_1, \mathbf{b}_1, W_2, \mathbf{b}_2, W_3, \mathbf{b}_3\}$ and the decoder parameters $\boldsymbol{\theta}_G = \{W_4, \mathbf{b}_4, W_5, \mathbf{b}_5, W_6, \mathbf{b}_6\}$. Figure 14 draws the network.

Information flows from left to right in Figure 14. The first layer (5.3a) forms 256 weighted sums of all pixels. Each row of W_1 is a pattern of pixels to which one unit responds, and the rectifier keeps only the positive responses. The second layer (5.3b) combines these 256 responses into 64 features of higher order. The third layer (5.3c) is linear and produces the latent vector $\mathbf{z}$, the only information about the image that is passed on. It has no activation function, so that $\mathbf{z}$ can take any real values. The last layer (5.3f) produces 784 values that the logistic function confines to the interval $(0, 1)$ of the pixel values.

A dense layer from n_{in} to n_{out} units has one weight for every pair of an input and an output unit and one bias for every output unit, $n_{\text{in}}n_{\text{out}} + n_{\text{out}}$ parameters. The first layer alone has $784 \cdot 256 + 256 = 200\,960$. Table 1 lists all layers for $d = 16$, which add up to 437 664 parameters. The

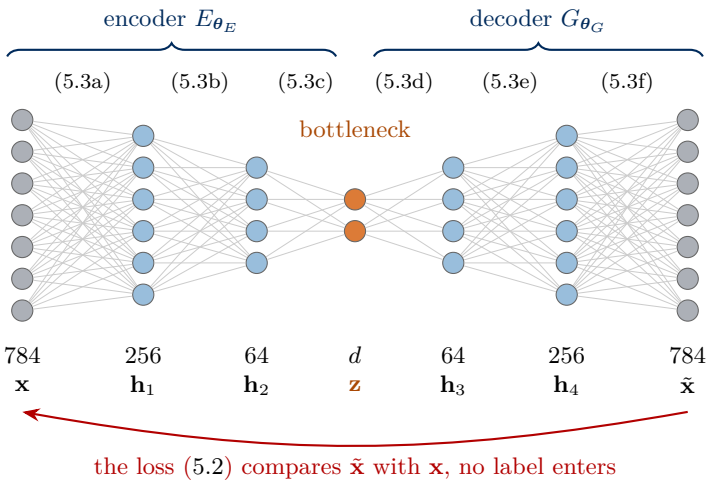


Figure 14: The dense autoencoder of equations (5.3). Numbers below the columns are the layer widths, from the input $\mathbf{x}$ through the hidden layers to the latent vector $\mathbf{z}$ (orange) and back to the reconstruction $\tilde{\mathbf{x}}$. The labels between the columns name the equation of each layer. Grey lines stand for weights, and only a few units per layer are drawn.

first and the last layer hold 92% of them, and the layers near the bottleneck are cheap. Training uses the Adam optimiser (adaptive moment estimation), a gradient descent with a separate step size for every parameter set by running averages of its gradient and squared gradient. The learning rate is 10^{-3}, the minibatches hold 256 images, and training runs for 15 epochs, passes over the 60 000 training images. Test errors are means over the 5000 test images of the fixed subset, which the network has not seen.

Table 1: Layers and parameters of the dense autoencoder of equations (5.3) with $d = 16$ latent numbers. ReLU and the logistic output function have no parameters.

part	equation	output width	parameters
encoder	(5.3a)	256	200 960
	(5.3b)	64	16 448
	(5.3c)	16	1 040
decoder	(5.3d)	64	1 088
	(5.3e)	256	16 640
	(5.3f)	784	201 488
total			437 664

After training, the encoder alone maps any image, also a new one, to its latent vector, a compressed representation for visualisation and clustering. The decoder alone maps a latent vector back to an image. Composed, the two reconstruct the input, and the reconstruction error $\varepsilon(\mathbf{x})$ measures how well a sample fits what the network has learned.

5.3 The linear autoencoder and the principal subspace

When all layers are linear the autoencoder has an analytic solution, and the solution connects it with PCA. Let the encoder and the decoder be affine maps, $\mathbf{z} = A\mathbf{x} + \mathbf{a}$ and $\tilde{\mathbf{x}} = B\mathbf{z} + \mathbf{b}$, with $A \in \mathbb{R}^{d \times D}$ and $B \in \mathbb{R}^{D \times d}$. The reconstruction is

$$\tilde{\mathbf{x}} = BA\,\mathbf{x} + \mathbf{c}, \qquad \mathbf{c} \coloneqq B\mathbf{a} + \mathbf{b}. \tag{5.4}$$

The constant $\mathbf{c}$ enters the loss as the intercept of a least-squares problem. Setting the derivative of $\sum_i \|\mathbf{x}_i - BA\mathbf{x}_i - \mathbf{c}\|^2$ with respect to $\mathbf{c}$ to zero gives $\mathbf{c} = \bar{\mathbf{x}} - BA\bar{\mathbf{x}}$, so that $\tilde{\mathbf{x}}_i - \bar{\mathbf{x}} = BA(\mathbf{x}_i - \bar{\mathbf{x}})$. The loss then depends only on the product $M \coloneqq BA$,

$$\mathcal{L} = \frac{1}{nD}\|X_c - X_c M^\top\|_F^2, \qquad \operatorname{rank} M \le d, \tag{5.5}$$

where X_c is the centred data matrix and $\|\cdot\|_F$ the Frobenius norm. The rank bound holds because M passes through the d-dimensional latent space. The matrix $X_c M^\top$ therefore has rank at most d. The Eckart–Young theorem (2.12) then bounds the training loss from below by $\frac{1}{D}\sum_{l>d}\sigma_l^2$, the PCA error (2.9) per pixel on the training data. The bound is reached when M is the orthogonal projector $\Lambda_d\Lambda_d^\top$ on the first d principal directions, and every factorisation of this projector,

$$A = Q\,\Lambda_d^\top, \qquad B = \Lambda_d\,Q^{-1}, \qquad Q \in \mathbb{R}^{d\times d} \text{ invertible}, \tag{5.6}$$

is a global minimum. All other stationary points of the loss are saddle points, so it has no other local minima [16, 17].

A linear autoencoder learns the same subspace as PCA, so an autoencoder can improve on PCA only through its nonlinearities (Figure 15). Because the matrix Q in equation (5.6) is arbitrary, the latent axes can be oblique and unordered by variance, mixing the principal directions. A nonlinear autoencoder has the same freedom, since any invertible transformation of the latent space can be absorbed into the encoder and the decoder. Its individual latent coordinates therefore carry no intrinsic meaning either. On the digits a linear autoencoder with $d = 2$ reaches the test error 0.0557, equal to the PCA test error at $d = 2$ (Table 2). Two planes through the origin are compared by their principal angles, which generalise the angle between two lines. The first is the smallest angle between a vector of one plane and a vector of the other. The second is found in the same way among the directions orthogonal to the first pair, and all angles vanish only for identical planes. The principal angles between the plane spanned by the columns of B and the plane of $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2$ are 0.76° and 0.97°. The two planes nearly coincide, although the columns themselves differ from the principal directions. The dense nonlinear autoencoders go below the PCA error (Table 2).

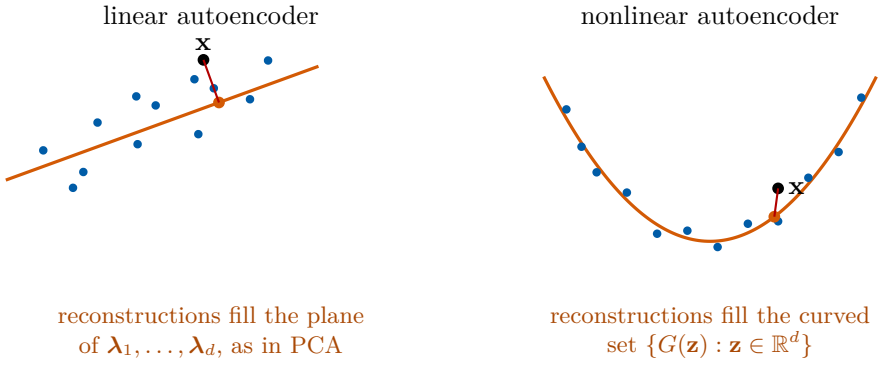


Figure 15: What the decoder can produce, sketch. Left, a linear autoencoder reconstructs every sample on the plane of the first d principal directions (orange line), and its optimum coincides with PCA. Right, a nonlinear decoder produces reconstructions on a curved set of dimension d, which can follow data that lie near a curved set. In both cases the reconstruction error (red) is the squared distance of a sample from the set of possible reconstructions, divided by D.

Table 2: Test reconstruction error per pixel, equation (5.2) on the 5000 test images, of PCA with d components and of the dense autoencoder with d latent numbers. The last column gives the PCA training error from the discarded eigenvalues, $\frac{1}{784}\sum_{l>d}\sigma_l^2$.

d	PCA	dense autoencoder	PCA, training, from the eigenvalues
2	0.0557	0.0389	0.0560
16	0.0268	0.0121	0.0273

5.4 Reconstructions and the latent space

With sixteen latent numbers, 49 times fewer than pixels, the reconstructions are blurred versions of the inputs with the right digit and shape (Figure 16). With two numbers, too few for ten digits and

their styles, several digits are rebuilt as another digit, a 3 as an 8 and a 4 as a 9. The error map of the digit that is rebuilt best (Figure 17) is almost blank. Its few remaining errors lie along the edges of the strokes, where the reconstruction is blurred, and its mean is the reconstruction error (5.1) of the image.

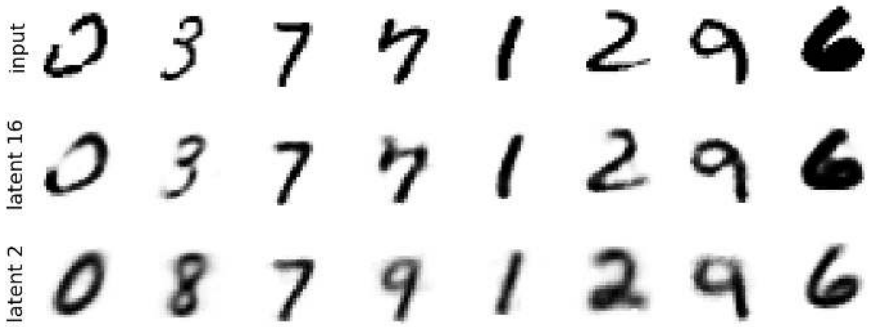


Figure 16: Eight MNIST test digits (top row) and their reconstructions $\tilde{\mathbf{x}}$ by the dense autoencoder with $d = 16$ (middle row) and $d = 2$ (bottom row) latent numbers.

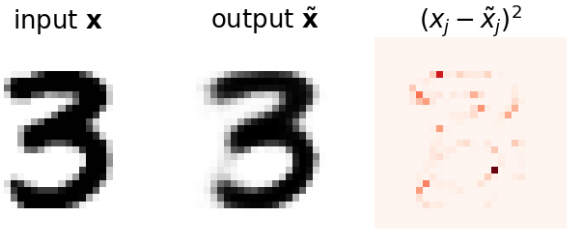


Figure 17: A test digit $\mathbf{x}$ (the 3 rebuilt best by the autoencoder with $d = 16$), its reconstruction $\tilde{\mathbf{x}}$, and the squared error $(x_j - \tilde{x}_j)^2$ of every pixel j. The error map is nearly white for zero and dark red for the largest value. The mean of the error map, 0.0042, is the reconstruction error $\varepsilon(\mathbf{x})$ of this image.

The latent vectors form a new data set for PCA, t-SNE and k-means (Figure 18). PCA of the sixteen latent numbers superposes the digits, as PCA of the pixels did, while their t-SNE map separates them into islands. k-means with $k = 10$ on the latent vectors agrees with the digits with ARI = 0.284, less than the 0.384 it reaches in the same run on 50 principal components of the pixels. For these components Section 4 reported 0.380 from a separate run with other random starts. On the t-SNE map of the latent vectors it reaches 0.696, slightly less than the 0.757 of the t-SNE map of the pixels in Section 4. For clustering the digits this latent space brings no gain over the pixels. The loss rewards a good reconstruction only, and no term in it favours round groups of similar size, the groups that k-means looks for.

A dense encoder treats an image as a flat list of 784 numbers and has no information about which pixels are neighbours. A convolutional encoder, built from the convolutional layers of a convolutional neural network (CNN), compares neighbouring pixels by construction. The convolutional autoencoder used here has two convolutions with 3×3 kernels and stride 2, which reduce the image from 28×28 to 14×14 and 7×7 pixels with 16 and 32 channels. Dense layers follow and lead to the latent space. The decoder mirrors these layers with transposed convolutions, which spread every input value over a 3×3 patch, with neighbouring patches shifted by two pixels. Each transposed convolution thus doubles the width and height of a feature map. The convolutional autoencoder has 413 315 parameters and was trained on 20 000 of the training images for 8 epochs, to keep the computing time small. The dense one saw all 60 000 images for 15 epochs. With $d = 2$ the latent space can be plotted as it is (Figure 19). With this smaller training budget the convolutional autoencoder reconstructs slightly worse, with the test error 0.0445 against 0.0389. In this single run it groups the digits slightly better, with ARI = 0.268 against 0.241 for k-means in the latent plane.

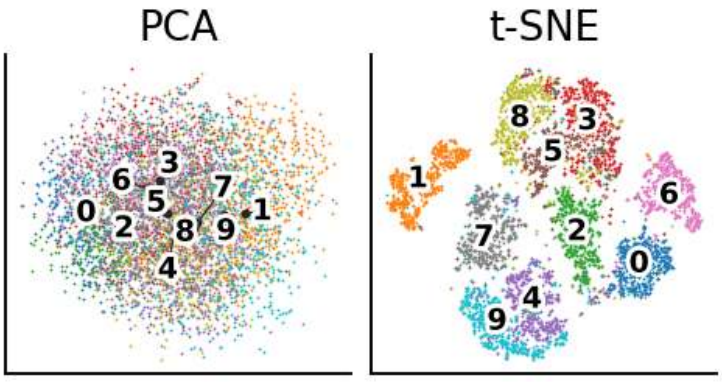


Figure 18: The sixteen latent numbers of the 5000 test digits (dense autoencoder, $d = 16$). Left, their projection on the first two principal directions of the latent vectors. Right, their t-SNE map. Colours show the true digits and are used for plotting only.

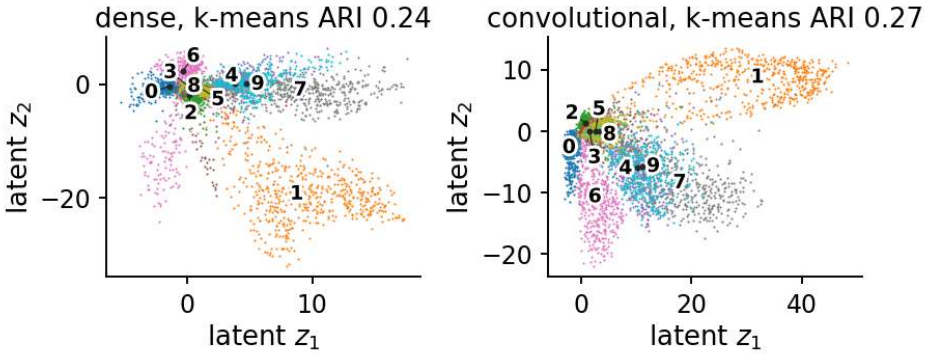


Figure 19: Latent vectors $\mathbf{z} = (z_1, z_2)$ of the 5000 test digits for autoencoders with $d = 2$, the dense one (left) and the convolutional one (right). Colours show the true digits, with each digit written at the median of its points, or next to it with a thin line where labels would overlap. The titles give the ARI of k-means with $k = 10$ applied to the latent vectors.

5.5 The latent space as a set of points

We write the loss (5.2) as a sum over the latent vectors of the training samples,

$$\mathcal{L} = \frac{1}{nD} \sum_{i=1}^{n} \left\| \mathbf{x}_i - G_{\boldsymbol{\theta}_G}(\mathbf{z}_i) \right\|^2, \qquad \mathbf{z}_i = E_{\boldsymbol{\theta}_E}(\mathbf{x}_i), \tag{5.7}$$

where the decoder enters only through its values at these n points. The encoder turns the n training samples into n isolated points in $\mathbb{R}^d$, and training adjusts the decoder so that each point is decoded into its own sample. Equation (5.7) refers to no other point of the latent space, so between the points the decoder is fixed only by continuity and by the form of the network. The location, scale and shape of the cloud of points are free as well. A different initialisation can give a differently shaped cloud with the same loss, for the reason shown by equation (5.6) in the linear case.

Beyond compression, this picture limits what an autoencoder can do (Figure 20). A point between the codes of two training images decodes into some image that the training never constrained to be a plausible sample. Drawing a random latent vector requires a distribution to draw from, and the autoencoder supplies none. The reconstruction error measures how well a sample can be rebuilt. This quantity is related to how typical the sample is, and it assigns no probability. A model that generates new samples and assigns probabilities needs latent vectors distributed according to a known density and a decoder trained on every region that this density uses. The variational autoencoder is a model of this kind.

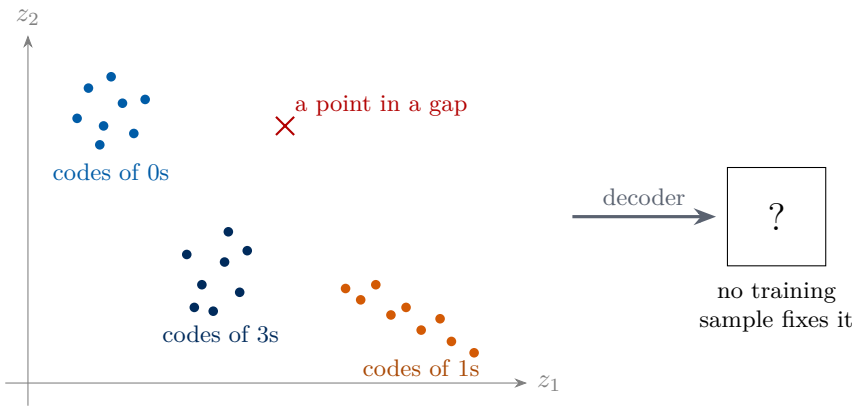


Figure 20: The latent space of an autoencoder as a set of points, sketch. Each training sample is mapped to one latent vector, and the latent vectors of similar samples cluster (blue, navy, orange). The loss (5.7) evaluates the decoder only at these points. A point in a gap between the clusters was never part of the training, and nothing determines what the decoder produces there.

6 Anomaly detection

6.1 The reconstruction error as a score

Anomaly detection looks for anomalies, samples from a region where $p(\mathbf{x})$ is small compared with the data we know. Anomalies are usually rare and varied, and a new kind has by definition never been seen. A useful method therefore learns what normal data look like and flags every sample that deviates from them. An autoencoder trained on normal samples only does this. Its decoder learns to produce samples like the training data, and a similar new sample is reconstructed close to itself. A sample unlike them is also mapped to some latent vector, where the training leaves the decoder output unconstrained (Section 5.5). In practice the output is an image resembling the training data (Figure 25 shows this for photographs), and the reconstruction tends to lie far from the input. The reconstruction error $\varepsilon(\mathbf{x})$ of equation (5.1) therefore serves as an anomaly score. A sample is flagged when its score exceeds a threshold τ fixed with normal data alone. One choice is the 95th percentile of their scores, which flags 5% of the normal samples, the rate of false alarms. In the experiments of this section the percentile is taken over the normal test samples, whose labels are known in this controlled study.

6.2 A subtle anomaly

We train the dense autoencoder with $d = 16$ on all training digits except the 7s, so that every 7 is an image of an unknown kind. Architecture and settings are those of Section 5, with 15 epochs. All 10 000 test digits are scored here, so that the about one thousand 7s give a smooth histogram. The mean error is 0.0128 for the trained classes and 0.0179 for the 7s, and the two distributions overlap strongly (Figure 21). The threshold $\tau = 0.0268$, which flags 5% of the normal digits, flags only 13.0% of the 7s. A 7 shares its strokes with the 1s and 9s, so the network reconstructs many 7s well although it has never seen one. Among the 100 test digits with the largest error 22 are 7s, against the 10.3% share of 7s in the test set, and among the 16 largest 3 are. The others are badly written digits of the trained classes, unusual in their own way (Figure 22).

6.3 The ROC curve and its area

The quality of a detector depends on the threshold, so comparing two scores requires a measure that is independent of it. Every threshold τ gives two rates. The true-positive rate $\mathrm{TPR}(\tau)$ is the fraction of anomalies with a score above τ, the anomalies found. The false-positive rate $\mathrm{FPR}(\tau)$ is the fraction of normal samples above τ, the false alarms. Lowering τ from above the largest score to below the smallest moves the point $(\mathrm{FPR}, \mathrm{TPR})$ from $(0, 0)$ to $(1, 1)$. The curve traced in this

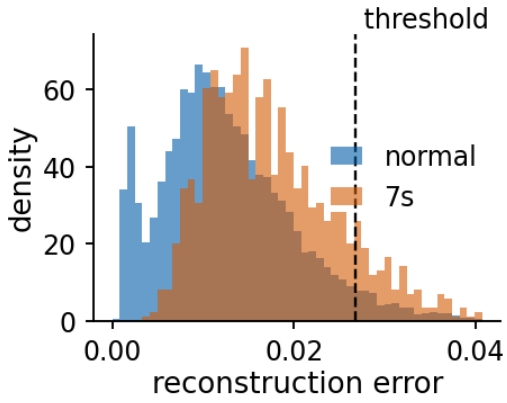


Figure 21: Distribution of the reconstruction error $\varepsilon(\mathbf{x})$ of the test digits for the autoencoder trained without the 7s, normalised as densities. Blue, the digits of the trained classes. Orange, the 7s. Dashed, the threshold that flags 5% of the trained digits.

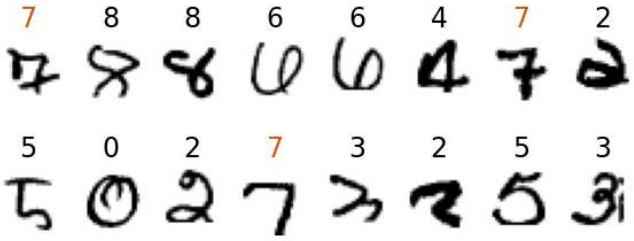


Figure 22: The 16 test digits with the largest reconstruction error for the autoencoder trained without the 7s, labelled by their true digit. Orange labels mark the 7s.

way is the receiver operating characteristic (ROC) curve [18]. A score that carries no information gives anomalies and normal samples the same distribution, so its curve is the diagonal. A score that separates them completely reaches TPR = 1 at FPR = 0.

The area under the curve (AUC) summarises all thresholds in one number. If $p_{\mathrm{A}}(a)$ and $p_{\mathrm{N}}(a)$ are the densities of the score a for anomalies and normal samples, then $\mathrm{TPR}(\tau) = \int_{\tau}^{\infty} p_{\mathrm{A}}\,\mathrm{d}a$ and $\mathrm{d\,FPR} = -p_{\mathrm{N}}(\tau)\,\mathrm{d}\tau$. As FPR runs from 0 to 1 the threshold runs from $+\infty$ down to $-\infty$. Reversing the limits absorbs the minus sign,

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}\,\mathrm{d\,FPR} = \int_{+\infty}^{-\infty} \mathrm{TPR}(\tau)\,(-p_{\mathrm{N}}(\tau))\,\mathrm{d}\tau = \int_{-\infty}^{\infty}\int_{\tau}^{\infty} p_{\mathrm{A}}(a)\,p_{\mathrm{N}}(\tau)\,\mathrm{d}a\,\mathrm{d}\tau = P(a_{\mathrm{A}} > a_{\mathrm{N}}), \tag{6.1}$$

which is the probability that a randomly chosen anomaly has a larger score than a randomly chosen normal sample [19]. It equals $\frac{1}{2}$ for an uninformative score and 1 for a score that separates the two classes completely. On a finite test set it is the fraction of all pairs of an anomaly and a normal sample that the score orders correctly. For the 7s, AUC = 0.696 (Figure 23), a weak signal above chance.

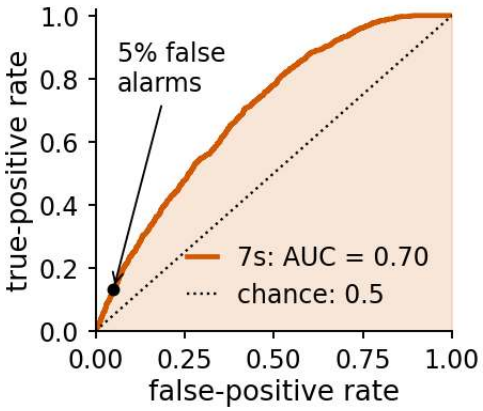


Figure 23: ROC curve of the reconstruction error for the 7s, autoencoder trained on all other digits. Horizontal axis, the false-positive rate (fraction of trained digits flagged). Vertical axis, the true-positive rate (fraction of 7s flagged). The shaded area is the AUC, the dot marks the threshold with 5% false alarms, and the dotted diagonal is the curve of an uninformative score.

6.4 A clear anomaly

At the other extreme lies an input unlike any digit. The autoencoder trained on all digits ($d = 16$) receives 5000 photographs of the CIFAR-10 collection [20], aeroplanes, cars, animals, ships and trucks, converted to grey values and reduced to 28×28 pixels. The median error is 0.0110 for the test digits and 0.2191 for the photographs (Figure 24). The threshold that flags 5% of the digits flags 99.8% of the photographs, and AUC = 0.9997. The reconstructions of the photographs look like blurred digits (Figure 25), since the decoder can produce nothing else. An anomaly far from the training data is easy to detect, and one that resembles the training data is hard.

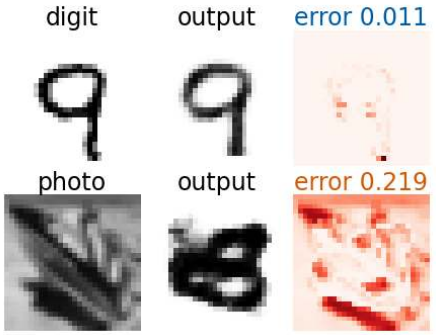


Figure 24: Autoencoder trained on all digits. Top, a test digit, its reconstruction and the squared error of every pixel. Bottom, the same for a photograph. Each is the median case of its group, and the two error maps share one colour scale.

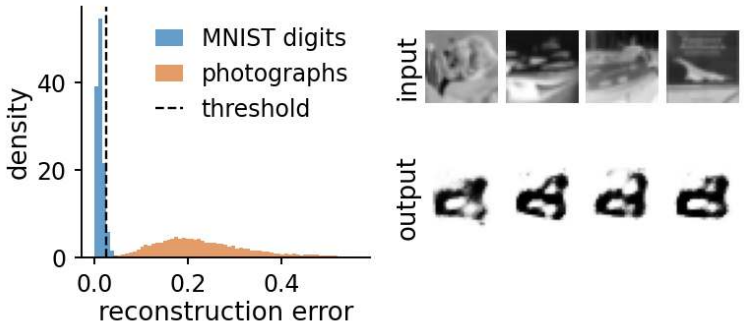


Figure 25: Left, distribution of the reconstruction error of the 10 000 MNIST test digits (blue) and of 5000 photographs (orange), normalised as densities. The dashed line marks the threshold that flags 5% of the digits. Right, four photographs (input) and their reconstructions (output).

7 Phases and phase diagrams

7.1 Thermal averages and their estimators

A classical spin model assigns an energy $H(\mathbf{s})$ to every configuration $\mathbf{s}$ of its spins. In thermal equilibrium at temperature T a configuration occurs with the Boltzmann probability

$$p(\mathbf{s}) = \frac{e^{-\beta H(\mathbf{s})}}{Z}, \qquad Z \coloneqq \sum_{\mathbf{s}} e^{-\beta H(\mathbf{s})}, \tag{7.1}$$

where the Boltzmann constant is $k_B = 1$ and $\beta \coloneqq 1/T$ is the inverse temperature. An observable $O(\mathbf{s})$ has the thermal average $\langle O \rangle \coloneqq \frac{1}{Z} \sum_{\mathbf{s}} O(\mathbf{s})\, e^{-\beta H(\mathbf{s})}$. The partition function Z also generates the thermodynamics. For N spins the energy per spin is $\langle e \rangle = \langle H \rangle / N = -\frac{1}{N} \partial_\beta \ln Z$. The specific heat per spin is proportional to the second derivative of $\ln Z$, which is the variance of the energy,

$$c = \frac{\beta^2}{N} \frac{\partial^2 \ln Z}{\partial \beta^2} = N\beta^2 \big(\langle e^2 \rangle - \langle e \rangle^2 \big). \tag{7.2}$$

The sum over all 2^N configurations of N Ising spins is out of reach for a lattice of useful size. Monte Carlo methods therefore draw n configurations $\mathbf{s}^{(1)}, \ldots, \mathbf{s}^{(n)}$ with probability $p(\mathbf{s})$ and estimate every thermal average by the sample mean $\overline{O} \coloneqq \frac{1}{n} \sum_k O(\mathbf{s}^{(k)})$. The Metropolis algorithm [21] proposes to flip one spin and accepts the flip with probability $\min(1, e^{-\beta \Delta E})$, where ΔE is the change of energy. Consecutive configurations of such a chain are correlated, so only nearly independent ones are stored. These snapshots, one array of spins each, are the input of the unsupervised methods.

7.2 Order parameters and transitions

A phase is a region of the control parameters in which the system behaves qualitatively the same way. An order parameter is a quantity that is nonzero in an ordered phase and zero in the disordered one. At a continuous, or second-order, transition the order parameter falls to zero without a jump. Its fluctuations and the correlation length ξ, the distance over which spins are correlated, diverge as powers of the reduced temperature $t \coloneqq (T - T_c)/T_c$. The exponents of these power laws are shared by all models with the same symmetry and dimension, which form a universality class. At a first-order transition the order parameter jumps and two phases coexist at the transition temperature (Figure 26).

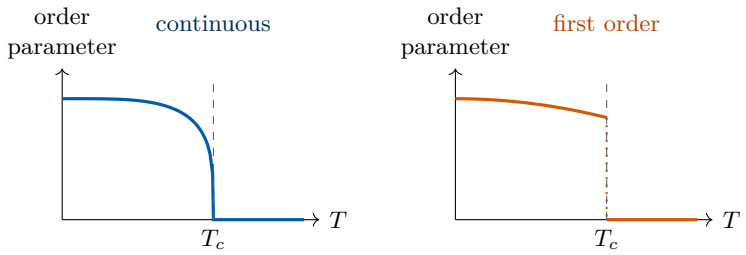


Figure 26: Order parameter against temperature, sketch. At a continuous transition (left) it falls to zero without a jump at T_c, at a first-order transition (right) it jumps to zero.

On a finite lattice all quantities stay finite and every transition is rounded into a smooth crossover, which finite-size scaling exploits. Near T_c an observable depends on T and on the linear size L of the lattice only through the ratio L/ξ, that is through $tL^{1/\nu}$. Here ν is the exponent of the correlation length, $\xi \propto |t|^{-\nu}$. Comparing several lattice sizes turns their smooth crossovers into a sharp statement about the infinite system.

7.3 Anomaly detection on a phase diagram

An autoencoder reconstructs data similar to its training data well and different data badly (Section 6). Its signal is weak for near anomalies such as the 7s and strong for far anomalies such as photographs. Snapshots change gradually near a continuous transition, like the first case, and differ strongly deep in another phase, like the second. Reference [22] turned this observation into a method for phase diagrams (Figure 27). The method was applied to ground-state data of the one-dimensional extended Bose–Hubbard model, a quantum model of bosons hopping on a chain with on-site and nearest-neighbour repulsion. An autoencoder is trained only on data from a small, well-understood region of one phase, and its reconstruction error is mapped over the whole diagram. The error stays small inside the trained phase and rises where the data stop resembling the training data. Phase boundaries are read off where it changes fastest, without labels and without an order parameter. We apply the same protocol to classical spin models whose physics is known and compare every output with the known answer.

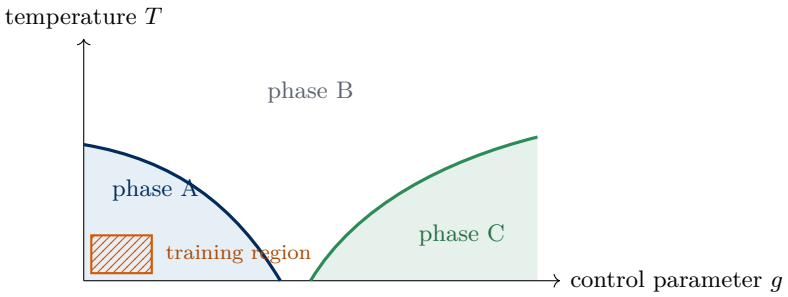


Figure 27: Anomaly detection on a phase diagram, sketch. An autoencoder is trained on snapshots from the hatched region of phase A only. Its reconstruction error is small inside phase A and larger elsewhere, and the line where it changes fastest marks the boundary of phase A. Thick lines, phase boundaries.

8 The two-dimensional Ising model

8.1 Hamiltonian and phases

The reference case is a model with one transition known from an exact solution. Spins $s_i = \pm 1$ sit on the sites of an $L \times L$ square lattice with periodic boundaries, $N = L^2$, and each spin interacts with its four nearest neighbours,

$$H(\mathbf{s}) = -J \sum_{\langle i,j \rangle} s_i s_j, \tag{8.1}$$

where $\langle i, j \rangle$ runs over the $2N$ nearest-neighbour bonds, each counted once. The coupling $J = 1$ sets the unit of energy (Figure 28). A satisfied bond, $s_i s_j = +1$, has energy $-J$, a broken one $+J$. The energy is unchanged when every spin is flipped, $H(-\mathbf{s}) = H(\mathbf{s})$. The order parameter is the magnetisation per spin, $m \coloneqq \frac{1}{N} \sum_i s_i$. In the ordered phase, $T < T_c$, most spins are parallel, all up or all down, and the symmetry $\mathbf{s} \to -\mathbf{s}$ is broken. In the disordered phase, $T > T_c$, the spins form small random domains and $m \to 0$ for $L \to \infty$. On a finite lattice without a field both signs are equally likely and $\langle m \rangle = 0$, and we use $\langle |m| \rangle$ instead. A magnetic field h adds the term $-h \sum_i s_i$ to H and favours one sign. Below T_c the magnetisation jumps from $-M$ to $+M$ when h crosses zero, with $M > 0$ the spontaneous magnetisation. In the (T, h) plane this jump forms a line of first-order transitions that ends at the critical point $(T_c, 0)$, and above T_c the magnetisation changes smoothly. All data below have $h = 0$ and scan the temperature.

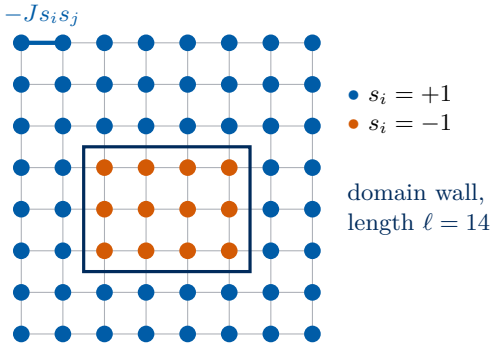


Figure 28: The Ising model, sketch. Spins $s_i = \pm 1$ on the sites of a square lattice (blue $+1$, orange -1), coupled along the nearest-neighbour bonds (grey lines) with energy $-Js_is_j$ per bond. A flipped domain is bounded by a closed domain wall (navy) that crosses ℓ broken bonds.

8.2 Domain walls and entropy

Equilibrium minimises the free energy $F = E - TS$, with E the energy and S the entropy, so order survives only if a large flipped domain raises F. In one dimension a wall between an up and a down region is one broken bond of energy $2J$ that can sit at any of $N - 1$ places. It changes the free energy

by $2J - T\ln(N-1)$, which for a long chain is negative at every $T > 0$, so walls proliferate and order is absent. In two dimensions a closed wall of length ℓ costs $2J\ell$. The number of its shapes grows at most as 3^{ℓ}, since each step of the wall can continue in at most three directions. The wall raises the free energy by about $\ell(2J - T\ln 3)$, so large domains cost free energy and order survives below $2J/\ln 3 = 1.820\,J$. This estimate, Peierls' argument [23], overcounts the shapes and therefore gives a lower bound for the transition temperature.

8.3 Exact results

Onsager solved the model exactly [24]. The transition is continuous and lies at

$$T_c = \frac{2J}{\ln(1+\sqrt{2})} = 2.2692\,J. \tag{8.2}$$

Yang computed the spontaneous magnetisation below T_c [25], $M(T) = \bigl(1 - \sinh^{-4}(2J/T)\bigr)^{1/8}$, so that $M \propto (T_c - T)^{1/8}$. The correlation length diverges as $\xi \propto |t|^{-\nu}$ with $\nu = 1$, the susceptibility as $|t|^{-7/4}$, and the specific heat logarithmically. These exponents define the two-dimensional Ising universality class. In Monte Carlo configurations of the 50×50 lattice (Figure 29) one sign dominates below T_c, domains of all sizes coexist near T_c, and the domains are small above T_c.

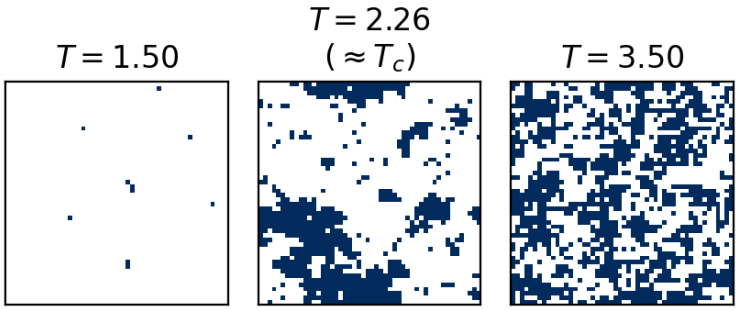


Figure 29: One Monte Carlo configuration of the 50×50 Ising lattice at $T = 1.50$, $T = 2.26$ (close to T_c) and $T = 3.50$. White and navy mark $s_i = -1$ and $s_i = +1$.

8.4 Monte Carlo data

Near T_c a spin flip inside a large correlated domain is almost always rejected. The Metropolis chain then needs very many sweeps, one attempted flip per spin, to turn a domain around. Consecutive samples are then nearly copies of each other, an effect called critical slowing down. The Swendsen–Wang algorithm [26] flips whole clusters of parallel spins at once, chosen at random in a way that preserves the Boltzmann distribution. For $L = 50$ at T_c the integrated autocorrelation time of $|m|$, roughly the number of steps between independent samples, is 26.4 sweeps for Metropolis and 3.4 cluster updates for Swendsen–Wang. The data set is generated with Swendsen–Wang updates in independent chains. For $L = 20$, 30 and 50 it contains 256 configurations at each of 37 temperatures between 1.5 and 3.5, in steps of 0.02 between 2.1 and 2.5 and of 0.1 elsewhere.

8.5 Finite-size scaling

A physicist extracts T_c from constructed observables of the order parameter. With m_k the magnetisation of sample k, the estimators of the magnetisation, the susceptibility and the Binder cumulant are

$$\overline{|m|} = \frac{1}{n}\sum_{k} |m_k|, \qquad \chi \coloneqq N\beta\bigl(\overline{m^2} - \overline{|m|}^{\,2}\bigr), \qquad U_4 \coloneqq 1 - \frac{\overline{m^4}}{3\,\overline{m^2}^{\,2}}, \tag{8.3}$$

estimates of $\langle|m|\rangle$, of $N\beta(\langle m^2\rangle - \langle|m|\rangle^2)$ and of $1 - \langle m^4\rangle/3\langle m^2\rangle^2$. The susceptibility measures the fluctuations of the order parameter and is built from $|m|$ because $\langle m\rangle = 0$ on a finite lattice. The Binder cumulant [27] measures the shape of the distribution of m. It tends to 2/3 in the ordered phase, where $m = \pm M$ is sharp, and to 0 in the disordered phase, where m is Gaussian and $\langle m^4\rangle = 3\langle m^2\rangle^2$. Being dimensionless, it takes the same value for every L at T_c, so the curves for different sizes cross there. The susceptibility peaks at a size-dependent temperature $T_c(L)$ with $T_c(L) - T_c \propto L^{-1/\nu}$, and extrapolation to $L \to \infty$ gives T_c.

In Figure 30 the Binder curves of $L = 20$ and 30 cross at $T = 2.2495$ and those of 30 and 50 at 2.2784. The susceptibility peaks at 2.3714, 2.3616 and 2.3031, and the fit $T_c(L) = T_c + aL^{-1}$ with the exact $\nu = 1$ gives $T_c = 2.2694$, against the exact 2.2692. Both observables were constructed with the knowledge that m is the order parameter.

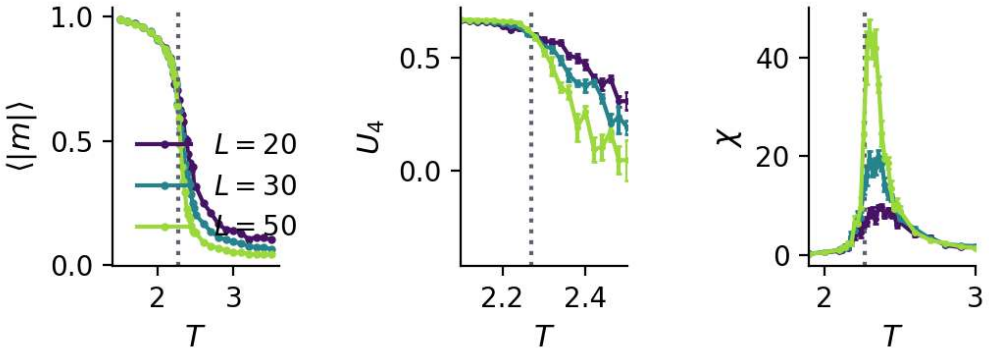


Figure 30: Observables of the order parameter for $L = 20$, 30 and 50. From left to right, the magnetisation $|m|$, the Binder cumulant U_4 and the susceptibility χ, each against the temperature T. Error bars come from blocks of independent chains. The dotted line marks the exact T_c.

8.6 PCA and the magnetisation

PCA receives the 37×256 configurations of the 30×30 lattice, each a vector $\mathbf{s}_i \in \{-1, +1\}^{900}$, with no temperature and no label. With the mean configuration $\bar{\mathbf{s}}$ and the principal directions $\boldsymbol{\lambda}_l$, each configuration has the coordinates $w_i^{(l)} = \boldsymbol{\lambda}_l^\top(\mathbf{s}_i - \bar{\mathbf{s}})$. At each temperature we average the absolute value of the first coordinate over the n_T configurations sampled there, $\langle|w^{(1)}|\rangle_T \coloneqq \frac{1}{n_T}\sum_{i:T_i=T}|w_i^{(1)}|$, and divide by $\sqrt{N}$.

The first component carries 37.3% of the variance and the second only 1.41%. The entries of $\sqrt{N}\boldsymbol{\lambda}_1$ are equal to 1 with a spread of 1% (Figure 31), and $|w^{(1)}|$ and $|m|$ have the correlation 0.99996 over all configurations. Below T_c all pairs of spins are correlated, $\langle s_i s_j\rangle \approx M^2$, the covariance matrix has a large uniform block, and its leading eigenvector is the uniform vector $\boldsymbol{\lambda}_1 \approx (1, \ldots, 1)/\sqrt{N}$. The projection on it is

$$w^{(1)} = \boldsymbol{\lambda}_1^\top(\mathbf{s} - \bar{\mathbf{s}}) \approx \frac{1}{\sqrt{N}}\sum_{i=1}^{N} s_i - \sqrt{N}\,\bar{m} = \sqrt{N}\,(m - \bar{m}), \tag{8.4}$$

with $\bar{m}$ the magnetisation averaged over the whole data set, negligible here. The Ising order parameter is linear in the spins, and a linear method finds it. Because $w^{(1)} \propto m$, the finite-size analysis can be repeated with $w^{(1)}$ in place of m. The Binder crossing of $L = 20$ and 30 is unchanged at four decimals, and that of 30 and 50 moves from 2.2784 to 2.2773.

8.7 The autoencoder signal

The autoencoder takes one configuration $\mathbf{s}$ as input. The N spins enter a dense layer of 128 units with ReLU, followed by a latent layer of 8 numbers. The decoder mirrors this structure and ends

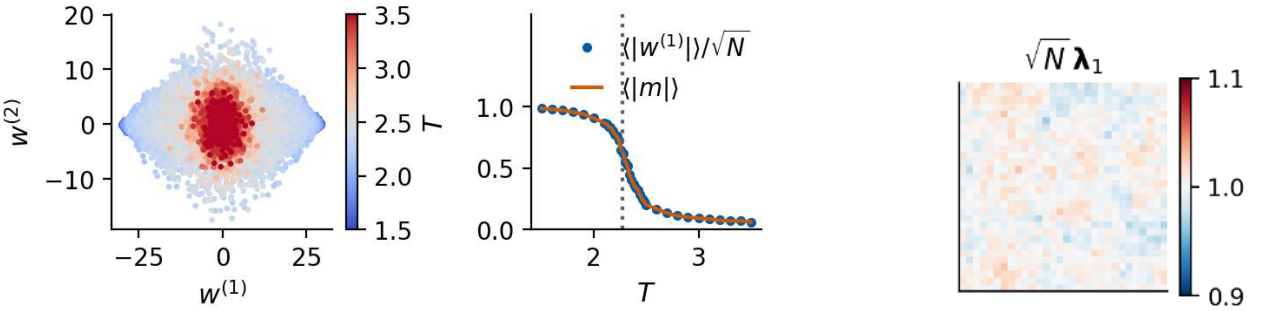


Figure 31: PCA of the Ising configurations of the 30×30 lattice. Left, the first two coordinates $(w^{(1)}, w^{(2)})$ of every configuration, coloured by its temperature T. Middle, $\langle|w^{(1)}|\rangle_T/\sqrt{N}$ (dots) and the magnetisation $\overline{|m|}$ (line) against T, with the exact T_c dotted. Right, the entries of the first principal direction multiplied by $\sqrt{N}$, shown on the lattice, with the colour scale from 0.9 to 1.1.

with tanh, so that $\tilde{s}_i \in [-1, 1]$. The reconstruction error of one configuration is

$$\varepsilon(\mathbf{s}) \coloneqq \frac{1}{N} \sum_{i=1}^{N} (s_i - \tilde{s}_i)^2, \tag{8.5}$$

and the same architecture is used for every L. The training data are the 768 configurations with $T \leq 1.7$, deep in the ordered phase, where Yang's formula gives $M(1.7) = 0.9701$. The network, given no temperature and no label, is trained with Adam for 200 epochs, three times with different random initialisations for each L. The mean error $\bar{\varepsilon}(T)$ over the configurations at temperature T rises like a smoothed step. We define its position by a fit on $1.9 \leq T \leq 3.0$,

$$\bar{\varepsilon}(T) = a + b \tanh\Big(\frac{T - T_{\mathrm{AE}}}{\delta}\Big), \tag{8.6}$$

whose results are the centre $T_{\mathrm{AE}}(L)$ of the step and its width $\delta(L)$. A fit uses all temperatures at once and is less sensitive to noise than the steepest slope between two grid points. Averaged over the three trainings, which differ by at most 0.002, the centre lies at $T_{\mathrm{AE}} = 2.3085$, 2.2767 and 2.2641 for $L = 20$, 30 and 50. It moves towards the exact $T_c = 2.2692$ as L grows and at $L = 50$ lies slightly below it (Figure 32). The width shrinks from 0.215 through 0.151 to 0.102, as $L^{-0.82}$, compared with the finite-size rounding $L^{-1/\nu} = L^{-1}$. With three sizes this power law is only indicative. The autoencoder signal is a property of the trained network, and no finite-size law has been derived for it, so it cannot be extrapolated like the susceptibility peak. It points, without an order parameter, to the region where a careful conventional analysis is worth doing.

The latent space shows what the network has learned. For this test a separate autoencoder is trained on all temperatures. A linear regression from its eight latent numbers to m reaches the coefficient of determination $R^2 = 0.9942$, the fraction of the variance of m explained by the regression. The latent vector therefore stores the magnetisation, to a good approximation linearly.

8.8 Dependence on the training region

A physical transition temperature is independent of the choice of training data. We therefore repeat the experiment with three training regions, deep in the ordered phase, $T \leq 1.7$, deep in the paramagnet, $T \geq 3.0$, and around T_c, $2.20 \leq T \leq 2.34$ (Figure 33). Trained on the ordered phase, the centres $T_{\mathrm{AE}} = 2.308$, 2.278, 2.264 for $L = 20$, 30, 50 move towards T_c. Trained on the paramagnet, the network sees nearly uncorrelated spins and learns little more than their mean. Its error is large everywhere, smaller in the ordered phase than on its own training data, since an ordered configuration is easier to approximate, and nearly flat for $L = 50$. The fitted centres 2.279, 2.253, 2.197 drift below T_c as L grows. Trained around T_c, the jump of the error sits at the upper edge of the training window, at 2.389, 2.363, 2.328. Because the detector measures how closely a snapshot resembles its training

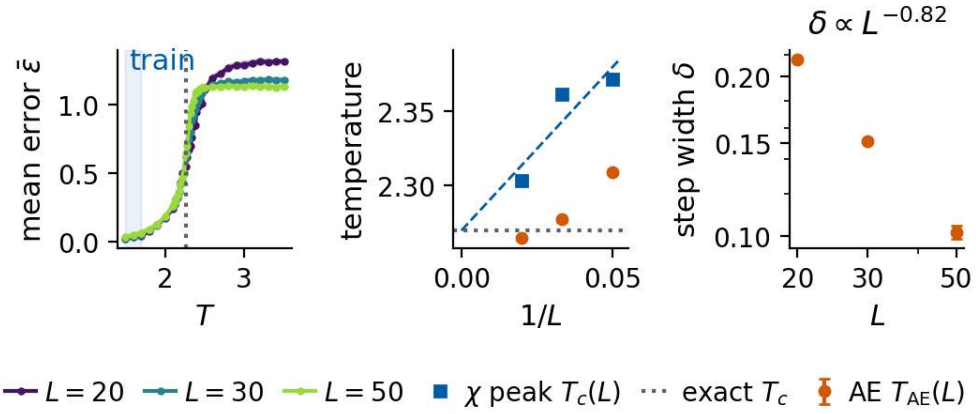


Figure 32: Autoencoder trained on the Ising configurations with $T \leq 1.7$ (shaded, "train"). Left, the mean reconstruction error $\bar{\varepsilon}(T)$ for $L = 20$, 30 and 50, with the exact T_c dotted. Middle, the centre $T_{\mathrm{AE}}(L)$ of the fitted step (orange) and the susceptibility peaks $T_c(L)$ (blue squares, with the line $T_c + a/L$) against $1/L$. Right, the width $\delta(L)$ of the step against L on logarithmic axes.

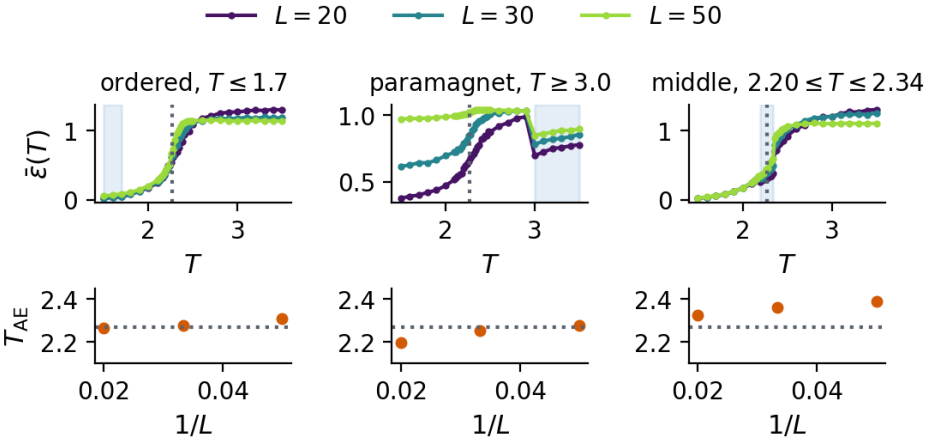


Figure 33: Three training regions (shaded) for the Ising autoencoder, the ordered phase $T \leq 1.7$, the paramagnet $T \geq 3.0$ and the window $2.20 \leq T \leq 2.34$ around T_c. Top row, the mean reconstruction error $\bar{\varepsilon}(T)$ for $L = 20$, 30, 50, with the exact T_c dotted. Bottom row, the centre T_{AE} of the fitted step against $1/L$.

data, its answer depends on that data, and the region to train on is a well-structured, well-understood phase.

For the Ising model both methods agree with the known physics, although the continuity of the transition and its exponents remain outside their reach. The model is easy because its order parameter is linear in the spins and known beforehand.

9 A frustrated magnet, the J_1–J_2 Ising model

9.1 Two competing couplings

The J_1–J_2 model adds a second coupling to the Ising model. Spins $s_i = \pm 1$ on an $L \times L$ square lattice with periodic boundaries interact with their four nearest neighbours and with their four diagonal neighbours,

$$H(\mathbf{s}) = -J_1 \sum_{\langle i,j \rangle} s_i s_j + J_2 \sum_{\langle\langle i,j \rangle\rangle} s_i s_j, \qquad g \coloneqq \frac{J_2}{J_1}, \tag{9.1}$$

where $\langle i,j \rangle$ runs over the $2N$ nearest-neighbour bonds and $\langle\langle i,j \rangle\rangle$ over the $2N$ diagonal bonds (Figure 34). The ferromagnetic coupling $J_1 > 0$ favours parallel nearest neighbours. The antiferromagnetic coupling $J_2 > 0$, with the plus sign in H, favours antiparallel diagonal neighbours. We set $J_1 = 1$, and

the phase diagram is the (g, T) plane. On one square of four spins the four nearest-neighbour bonds favour parallel spins and the two diagonal bonds favour antiparallel diagonal pairs. No configuration satisfies all six bonds, a competition called frustration.

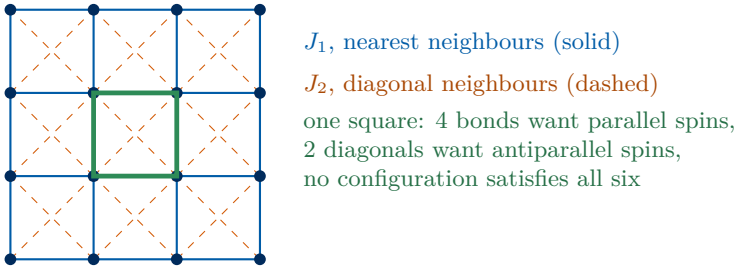


Figure 34: The J_1–J_2 model, sketch. Spins (dots) on a square lattice, ferromagnetic nearest-neighbour bonds J_1 (blue, solid) and antiferromagnetic diagonal bonds J_2 (orange, dashed). The green square marks one plaquette, on which the couplings compete.

9.2 Ground states

At $T = 0$ the system sits in the configuration of lowest energy. Each spin has two nearest-neighbour and two diagonal bonds when every bond is counted once, so the energy per spin of a pattern follows from counting bonds. The ferromagnet satisfies all nearest-neighbour bonds and frustrates all diagonals, $E/N = -2J_1 + 2J_2$. The stripe pattern, columns (or rows) of alternating sign, has one satisfied and one broken nearest-neighbour bond per spin and satisfies all diagonals, $E/N = -2J_2$. The Néel pattern, a chessboard, breaks all nearest-neighbour bonds, $E/N = 2J_1 + 2J_2$, and lies above the ferromagnet in energy for every $J_1 > 0$. The ferromagnet and the stripes have equal energy at $g = 1/2$. The ground state is therefore ferromagnetic for $g < 1/2$, with two states, and striped for $g > 1/2$, with four states, vertical or horizontal stripes and their global flips. At $g = 1/2$ rows or columns can be flipped at no cost. An enumeration of all 2^{16} configurations of a 4×4 torus confirms the picture and finds 30 ground states at $g = 1/2$, many more than $2 + 4$ (Figure 35).

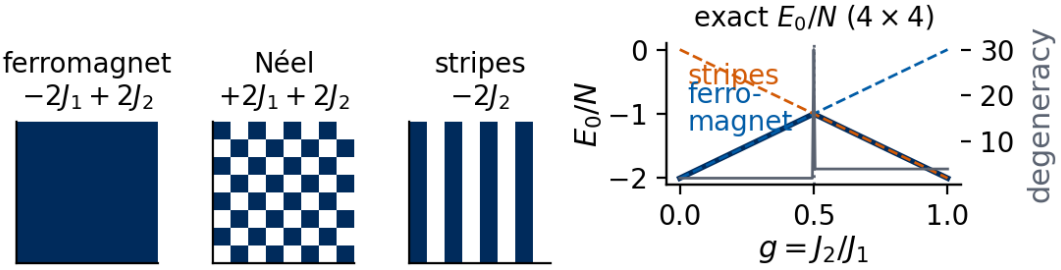


Figure 35: Ground states of the J_1–J_2 model. Left, the ferromagnetic, Néel and stripe patterns with their energies per spin (navy and white, the two spin values). Right, the exact ground-state energy per spin E_0/N of the 4×4 torus against $g = J_2/J_1$ (black) and the energies of the ferromagnet (blue) and of the stripes (orange). The grey curve (right axis) is the number of ground states, which jumps at $g = 1/2$.

9.3 Order parameters from Fourier components

Ordering patterns that repeat with a wave vector $\mathbf{k}$ are measured by the Fourier component of the spins,

$$m_{\mathbf{k}} \coloneqq \frac{1}{N} \sum_{\mathbf{r}} s_{\mathbf{r}}\, e^{i\mathbf{k}\cdot\mathbf{r}}, \tag{9.2}$$

where $\mathbf{r} = (x, y)$ labels the sites. At $\mathbf{k} = (0, 0)$ all phase factors are 1 and $m_{(0,0)} = m$ is the magnetisation. At $\mathbf{k} = (\pi, 0)$ the factor $e^{i\pi x} = (-1)^x$ flips the sign of every second column. For

vertical stripes, $s_{\mathbf{r}} = \pm(-1)^x$, all N products $(-1)^x s_{\mathbf{r}}$ have the same sign, so $m_{(\pi,0)} = \pm 1$ and $m = 0$. Horizontal stripes give $m_{(0,\pi)} = \pm 1$. The stripe order parameter

$$m_{\mathrm{str}} \coloneqq \sqrt{m_{(\pi,0)}^2 + m_{(0,\pi)}^2} \tag{9.3}$$

combines both orientations. It equals 1 for ideal stripes of either kind and nearly 0 in the ferromagnet and the paramagnet. Since the magnetisation vanishes in the ordered stripe phase, a scan of $\langle |m| \rangle$ alone shows the stripe phase as a paramagnet.

9.4 Domain walls and the phase diagram

The ground states determine what orders, and the domain walls, as in Peierls' argument, determine up to which temperature the order survives. A straight wall in the ferromagnet breaks one nearest-neighbour bond per unit length, $+2J_1$. It also satisfies the two diagonal bonds that cross it, $-2J_2$ each, so it costs $2J_1(1 - 2g)$ per unit length. A wall in the stripe phase parallel to the stripes costs $2J_1(2g - 1)$. Both vanish at $g = 1/2$, where walls appear at any $T > 0$ and destroy the order, and the transition temperature drops to zero.

Two limits are exact. At $g = 0$ the model is the Ising model with $T_c = 2.269\, J_1$. For $g \to \infty$ only the diagonal bonds remain, which connect sites of the same chessboard colour. The lattice then splits into two independent antiferromagnetic Ising lattices with coupling J_2, each ordering at $T_c \to 2.269\, J_2$. Between these limits the transition line was mapped by decades of work [28, 29, 30]. For $g < 1/2$ the transition is continuous and in the two-dimensional Ising universality class, with $T_c(g)$ falling to zero at $g = 1/2$. For $g > 1/2$ the transition into the stripe phase is continuous with exponents that vary along the line well above $g = 1/2$, while close to $g = 1/2$ its nature is debated. A double-peaked energy histogram, the signature of a first-order transition, was observed at $g = 0.6$ in Ref. [29]. Reference [30] placed the end of a weakly first-order region at $g^* \approx 0.67$. Tensor-network calculations find a lower value, $g^* \simeq 0.54$, and so a smaller first-order region [31]. The transition temperatures of the phase diagram in Figure 36 come from transfer matrices of infinite strips. These give the free energy and the correlation length ξ_L of a strip of width L numerically exactly. Mean-field theory, which replaces the neighbours of a spin by their average, predicts the lines $T = 4(1 - g)$ and $T = 4g$, which meet at $T = 2$ for $g = 1/2$. It overestimates the transition temperature everywhere and misses the drop to zero at $g = 1/2$, where fluctuations between the many degenerate states destroy the order.

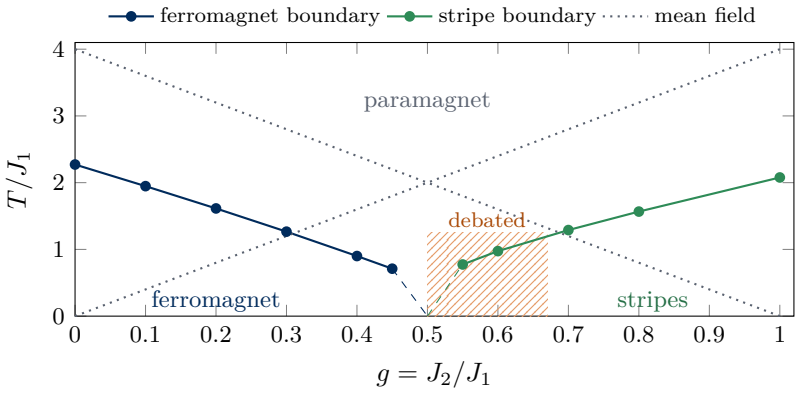


Figure 36: Phase diagram of the J_1–J_2 model in the (g, T) plane. Dots, transition temperatures from the crossing of the scaled correlation lengths ξ_L/L of strips of widths $L = 8$ and $L = 10$, joined by solid lines as a guide to the eye. Dashed segments connect them to $T_c = 0$ at $g = 1/2$. Below the left line the ferromagnet has $m \neq 0$, below the right line the stripe phase has $m_{\mathrm{str}} \neq 0$, and above both lies the paramagnet. Hatched, the region $1/2 < g \lesssim 0.67$ in which the nature of the transition is debated. Dotted, the mean-field lines $T = 4(1 - g)$ and $T = 4g$.

9.5 Choice of observable

The data are Metropolis snapshots of the 32×32 lattice on a grid of 81 values of g between 0 and 1 and 79 temperatures between 0.1 and 4. Each grid point holds 100 snapshots, 639 900 in total (Figure 37). Since diagonal neighbours have the same chessboard colour, we divide the lattice into four sublattices labelled by the parities of x and y. Spins of one sublattice are mutually uncoupled and can be updated simultaneously. Each chain is cooled from high temperature, which helps it find the ordered state at low T.

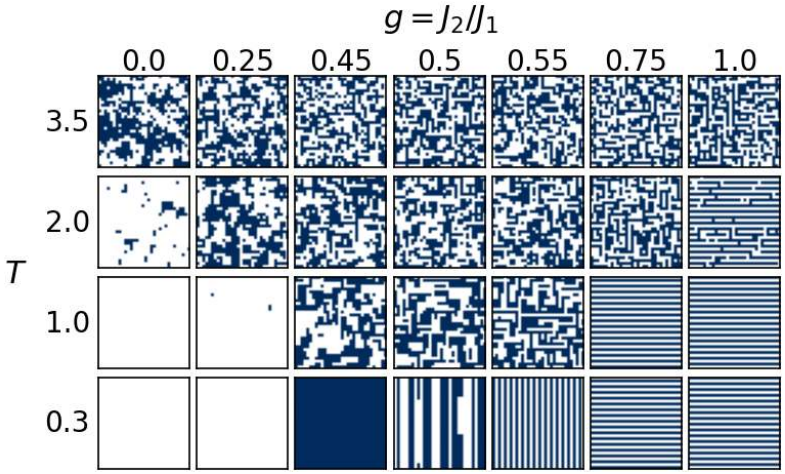


Figure 37: Monte Carlo snapshots of the 32×32 lattice (white and navy, the two spin values) at $g = J_2/J_1$ from 0 to 1 (columns) and T from 0.3 to 3.5 (rows).

A physicist without knowledge of stripes would measure the magnetisation at every grid point (Figure 38). The map of $\overline{|m|}$ shows one ordered region, at $g < 1/2$, and the stripe phase looks like a paramagnet. The other ordered region appears in the map of $\overline{m_{\text{str}}}$, only because we knew which pattern to measure. The specific heat (7.2), estimated as $N\beta^2(\overline{e^2} - \overline{e}^{\,2})$, peaks at every transition whatever the order and traces both boundaries without any order parameter. Its estimate is noisy, since a variance needs many more samples than a mean (Figure 39). With 10 samples per grid point five independent estimates of the specific heat differ strongly near the peak, in height and in shape. A reliable peak needs of order a hundred decorrelated samples at every one of the 6399 grid points. In an experiment every grid point is a separate preparation of the sample.

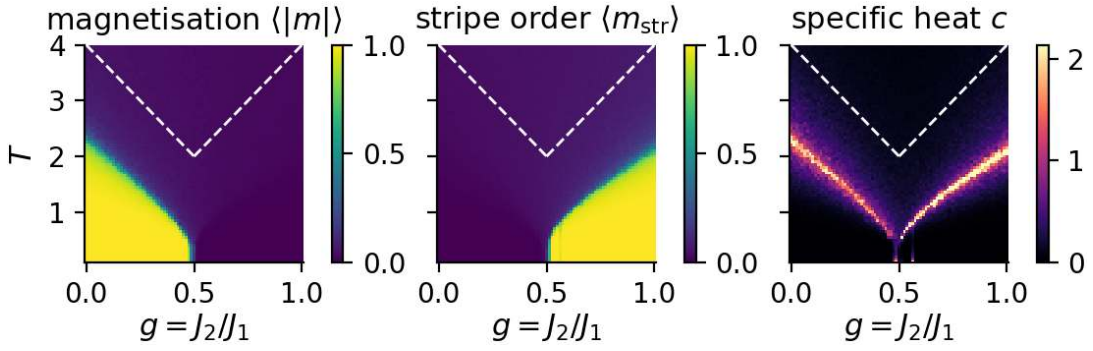


Figure 38: Three maps of the (g, T) plane from the same snapshots. Left, the magnetisation $\overline{|m|}$. Middle, the stripe order parameter $\overline{m_{\text{str}}}$. Right, the specific heat c. Dashed white lines, the mean-field transition lines.

9.6 PCA and the two order parameters

PCA receives 20 snapshots from every grid point, 127 980 in total, with no information about g or T. The first three principal directions carry 16.6%, 8.4% and 8.2% of the variance and all others at most 0.3% (Figure 40). We compare the directions with the candidate patterns, the uniform

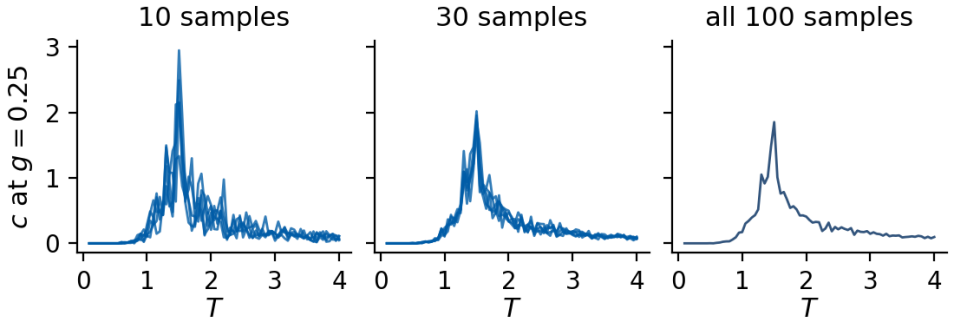


Figure 39: The specific heat c at $g = 0.25$ against T, estimated from subsets of 10 and 30 samples per temperature (several repetitions, each a line) and from all 100 samples.

vector and the two stripe patterns normalised to unit length. The overlaps, 1.000 to three decimals, show that $\boldsymbol{\lambda}_1$ is the uniform pattern and $\boldsymbol{\lambda}_2$ and $\boldsymbol{\lambda}_3$ are the two stripe orientations. Projecting every snapshot gives two order-parameter candidates, $|w^{(1)}|/\sqrt{N} \approx |m|$ from the uniform direction and $\sqrt{(w^{(2)})^2 + (w^{(3)})^2}/\sqrt{N} \approx m_{\mathrm{str}}$ from the two stripe directions. Their maps over the plane correlate with the maps of $\overline{|m|}$ and $\overline{m_{\mathrm{str}}}$ with the coefficients 1.0000 and 0.9999. PCA has found the stripe order parameter, which the magnetisation scan missed, without being told that it exists. It succeeds because both order parameters are linear in the spins, $m_{\mathbf{k}} = \frac{1}{N}\sum_{\mathbf{r}} s_{\mathbf{r}} e^{i\mathbf{k}\cdot\mathbf{r}}$, and order that is nonlinear in the spins is out of its reach. The different variances of the two stripe directions are a property of the data. Each cooled chain picks one orientation when it orders and keeps it, and a finite number of chains does not sample the two orientations equally.

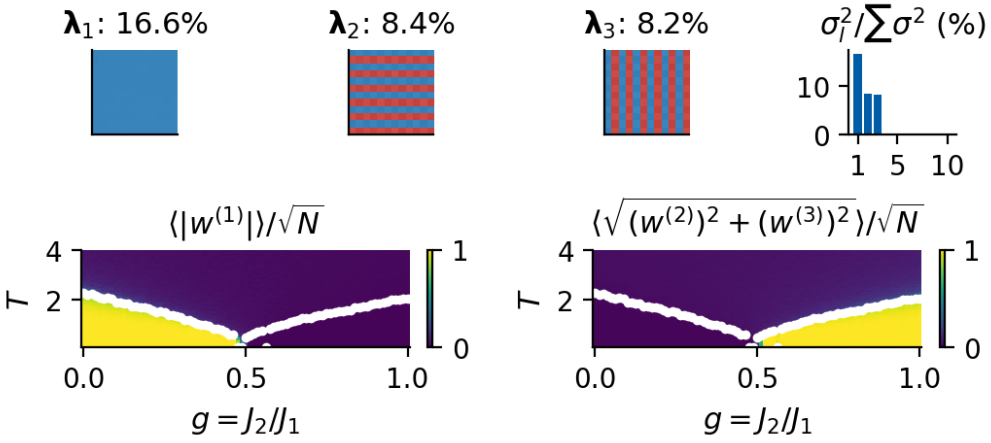


Figure 40: PCA of the J_1–J_2 snapshots. Top, the first three principal directions $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_2$, $\boldsymbol{\lambda}_3$ shown on the lattice (red and blue, the two signs of the entries) with their fractions of the variance. The fractions $\sigma_l^2/\sum\sigma^2$ of the first ten directions are shown in per cent. Bottom, the maps of $\langle|w^{(1)}|\rangle/\sqrt{N}$ and $\langle\sqrt{(w^{(2)})^2 + (w^{(3)})^2}\rangle/\sqrt{N}$ over the (g, T) plane, averaged over the snapshots of each grid point. White dots, the maximum of the specific heat for every g.

9.7 Three training regions

The autoencoder for this model is a small convolutional network with periodic boundaries, like the lattice. Two convolutions with stride 2 reduce the lattice from 32×32 to 16×16 and 8×8, a dense bottleneck holds 4 numbers, and the decoder mirrors the encoder and ends with tanh. The energy is unchanged when all spins flip, whereas a cooled chain orders into one of the two signs and stays there. Every training snapshot is therefore added also with all spins flipped. The autoencoder is trained on one corner of the plane, and its mean reconstruction error $\bar{\varepsilon}$ is mapped over the whole plane. For every g the boundary point is the temperature where $\log\bar{\varepsilon}$ changes fastest, drawn only if that change is at least 15% of the full range of the map. The three training regions are regime A in

the ferromagnet ($g \leq 0.2$, $T \leq 1.0$), regime B in the paramagnet ($g \leq 0.2$, $T \geq 3.5$) and regime C in the stripe phase ($g \geq 0.8$, $T \leq 1.0$).

In the three maps of Figure 41, regime A reconstructs the ferromagnet with a very small error, and its boundary for $g < 1/2$ is an abrupt edge falling towards $g = 1/2$. The stripes and the paramagnet both differ from the ferromagnet and have similar large errors, so this regime cannot separate them. Regime B, trained on nearly uncorrelated spins, learns little more than the mean, and its error is large everywhere. It reconstructs ferromagnetic snapshots better than its own training data, so the ferromagnetic boundary appears as a drop of the error. The stripe phase has the same error as the paramagnet and stays invisible. Regime C reconstructs the stripes and finds their boundary for $g > 1/2$. Where a regime detects a boundary, it lies a median distance of 0.025 in temperature from the maximum of the specific heat, half the grid spacing $\Delta T = 0.05$. No regime is reliable close to $g = 1/2$, where no phase is fully developed within the temperature range of the data. An autoencoder trained on one region measures only resemblance to that region and cannot tell two different anomalies apart.

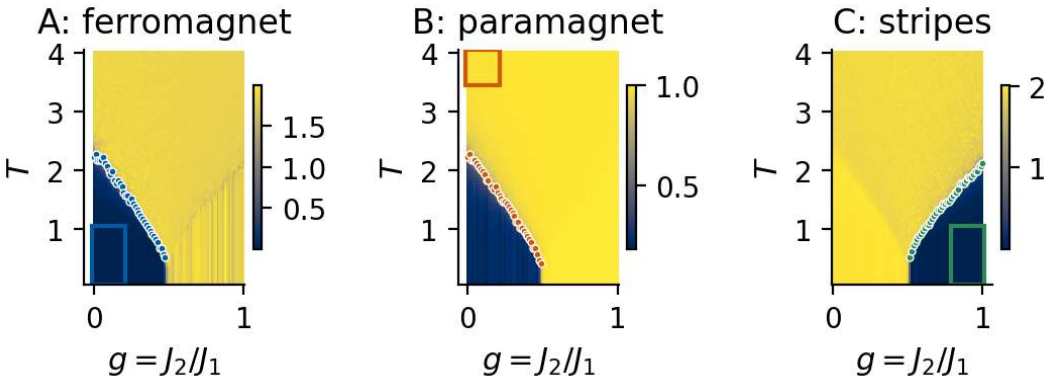


Figure 41: Mean reconstruction error $\bar{\varepsilon}$ over the (g, T) plane for the autoencoders of regime A (ferromagnet), B (paramagnet) and C (stripes), each with its own colour scale. Rectangles mark the training regions and dots the detected boundary points.

9.8 A combined map and a coarse scan

Since each regime recognises only its own phase, we combine them. A trained autoencoder defines a normal level of its error on its own training region. For a grid point and regime $\rho \in \{\mathrm{A}, \mathrm{B}, \mathrm{C}\}$ the anomaly score

$$a_\rho(g, T) := \frac{|\bar{\varepsilon}_\rho(g, T) - \bar{\varepsilon}_\rho^{\,\mathrm{tr}}|}{\mathrm{sd}_\rho} \tag{9.4}$$

measures how far the mean error lies from that level in units of its spread. Here $\bar{\varepsilon}_\rho^{\,\mathrm{tr}}$ and sd_ρ are the mean and standard deviation of $\bar{\varepsilon}_\rho$ over the grid points of the training region. The absolute value matters, since regime B reconstructs ferromagnetic snapshots better than its training data, which is also an anomaly. A grid point is assigned to the regime with the smallest score if that score is below 3. Otherwise it stays unassigned, since it resembles none of the three training regions. The three colours of Figure 42 match the three phases, with 9.9% of the grid assigned to the ferromagnet, 59.2% to the paramagnet, 10.1% to the stripes and 20.7% unassigned. The unassigned band lies mainly on the ordered side of each transition line. There thermal fluctuations (flipped islands and domain walls) make the snapshots atypical for a model trained deep in the phase. Around $g = 1/2$ at the lowest temperatures the points are assigned to the paramagnet, consistent with $T_c \to 0$ at $g = 1/2$. The map used no order parameter and depends on the choice of reference regions, since regime B alone would have missed the stripe phase.

These maps used the full grid. If only a coarse scan can be afforded, we take every eighth value of g and every eighth temperature with 10 snapshots per point, 1100 snapshots. Regimes A and C are

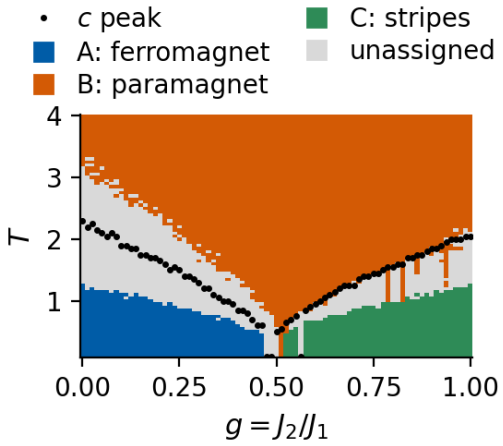


Figure 42: Combined map. Every grid point of the (g, T) plane is coloured by the regime with the smallest anomaly score $a_\rho < 3$, or grey if unassigned. Black dots, the maximum of the specific heat for every g.

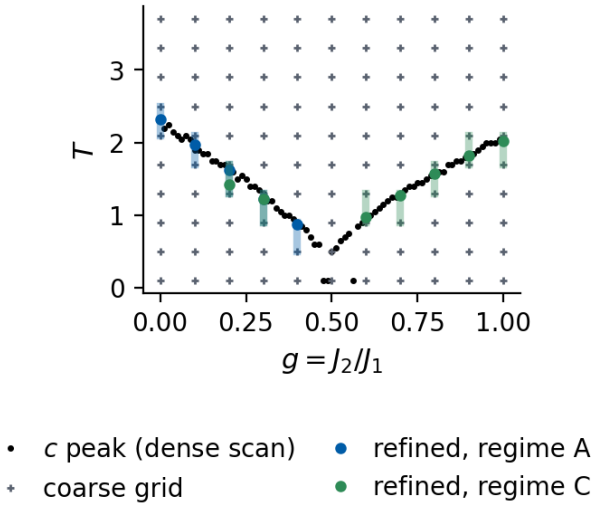


Figure 43: Coarse scan. Crosses, the coarse grid ($\Delta g = 0.1$, $\Delta T = 0.4$, 10 snapshots per point). Bars, the temperature brackets in which the error of regimes A (blue) and C (green), trained on the coarse data, jumps. Large dots, the boundary after dense measurements inside the brackets only. Small black dots, the specific-heat maximum of the dense scan.

trained on the coarse data alone. The temperature brackets in which their error jumps mark where a dense measurement is worth paying for. Measuring densely only inside these brackets costs 2160 further snapshots, and the resulting boundary points land on the specific-heat ridge of the dense scan (Figure 43). The total, 3260 snapshots, is 0.5% of the 639 900 snapshots of the dense specific-heat map. The method is useful in this mode, as a cheap survey of raw snapshots that locates changes, followed by careful measurements with the physicist's tools.

The methods left open which order sets in and whether a transition is continuous or first order. The energy histogram at the transition settles the second point, with one peak for a continuous and two for a first-order transition. Near g^* this test needs lattices much larger than those used here. In both the Ising and the J_1–J_2 model every order parameter was linear in the spins.

10 The two-dimensional XY model

10.1 Planar spins

In the XY model every site of an $L \times L$ square lattice with periodic boundaries carries a planar unit vector $\mathbf{s}_i = (\cos\theta_i, \sin\theta_i)$ given by an angle $\theta_i \in [0, 2\pi)$. Nearest neighbours interact through the cosine of their angle difference,

$$H(\boldsymbol{\theta}) = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j) = -J \sum_{\langle i,j \rangle} \mathbf{s}_i \cdot \mathbf{s}_j, \tag{10.1}$$

with $J = 1$. The energy is unchanged by a rotation of all spins by the same angle, $\theta_i \to \theta_i + \alpha$, a continuous symmetry, whereas the Ising model has the discrete flip. The magnetisation $\mathbf{m} \coloneqq \frac{1}{N}\sum_i \mathbf{s}_i$ is a two-component vector, and the thermal averages become integrals over all angles, $Z = \int \mathrm{d}\boldsymbol{\theta}\, e^{-\beta H}$.

10.2 Spin waves and the magnetisation

At low temperature neighbouring angles differ little, $\cos(\theta_i - \theta_j) \approx 1 - (\theta_i - \theta_j)^2/2$, and the energy becomes a sum of independent harmonic modes, the spin waves. A twist of the angle by 2π spread over a region of linear size ℓ changes each angle difference by $2\pi/\ell$ (Figure 44). The region contains of

order ℓ^2 bonds, so the twist costs an energy of order $J\ell^2(2\pi/\ell)^2$, independent of ℓ in two dimensions. Twists therefore appear on all scales at any $T > 0$. In the harmonic approximation the angle at one site is Gaussian with a variance that grows as $\frac{T}{2\pi J}\ln L$. Since $\langle\cos\theta\rangle = e^{-\langle\theta^2\rangle/2}$ for a Gaussian angle of zero mean, the magnetisation of a finite lattice decays as a small power of its size, $\langle|\mathbf{m}|\rangle \propto L^{-T/4\pi J}$. Monte Carlo at $T = 0.3$ gives $\langle|\mathbf{m}|\rangle = 0.940$, 0.907 and 0.874 for $L = 8$, 32 and 128, a power law with the fitted exponent 0.0261, close to the spin-wave value $T/4\pi J = 0.0239$. The magnetisation vanishes for $L \to \infty$ at every $T > 0$, as the Mermin–Wagner theorem states for continuous symmetries in two dimensions [32]. On any lattice of practical size it is nevertheless large at low T.



Figure 44: A slow twist of planar spins, one row of the twisted region, sketch. Each spin is rotated by $2\pi/\ell$ relative to its neighbour, so each bond costs little energy, and the total cost of the twist is independent of ℓ in two dimensions.

10.3 Vortices and the Berezinskii–Kosterlitz–Thouless transition

The model nevertheless has a transition, driven by vortices. A vortex is a point around which the spin direction turns once by $\pm 2\pi$ (Figure 45). Its winding number

$$w \coloneqq \frac{1}{2\pi}\oint_{\mathcal{C}} \nabla\theta \cdot \mathrm{d}\mathbf{l} \tag{10.2}$$

counts the turns of the spins along a closed path $\mathcal{C}$ around it, $+1$ for a vortex and -1 for an antivortex. On the lattice, w_r of an elementary square, a plaquette, is the sum of the four angle differences around it, each mapped to $[-\pi, \pi)$, divided by 2π. At distance r from a vortex $|\nabla\theta| = 1/r$. The continuum form of the harmonic energy, with the lattice spacing a as the short-distance cutoff, gives

$$E_v = \frac{J}{2}\int |\nabla\theta|^2 \,\mathrm{d}^2 r = \frac{J}{2}\int_a^L \frac{2\pi r\,\mathrm{d}r}{r^2} = \pi J \ln(L/a), \tag{10.3}$$

which grows with the size of the lattice. A free vortex can sit on any of about $(L/a)^2$ sites, so its entropy is $2\ln(L/a)$ and its free energy $F_v = (\pi J - 2T)\ln(L/a)$. Below $T = \pi J/2 = 1.571\,J$, where the coefficient changes sign, free vortices cost an infinite free energy and occur only in bound vortex–antivortex pairs, whose energy stays finite. Above it the entropy dominates and the pairs unbind. The bound pairs screen the interaction and lower the actual transition temperature to $T_{\mathrm{BKT}} = 0.8929\,J$, the value of high-precision Monte Carlo [37]. The unbinding of vortex pairs is the Berezinskii–Kosterlitz–Thouless (BKT) transition [33, 34, 35]. Below T_{BKT} the spin correlation $G(r) = \langle\cos(\theta_0 - \theta_r)\rangle$ decays as a power law, called quasi-long-range order. Above it free vortices scramble the directions, and $G(r)$ decays exponentially with a correlation length ξ. As T_{BKT} is approached from above, the correlation length diverges as $\exp(b/\sqrt{T - T_{\mathrm{BKT}}})$ with a nonuniversal constant b. The transition has no order parameter and breaks no symmetry. It is of infinite order, with all derivatives of the free energy continuous, and the specific heat has only a smooth maximum above T_{BKT}.

The low-temperature phase resists a twist. Its helicity modulus, or spin stiffness, Υ is the free-energy cost of a small total twist Δ imposed on all spins across the sample, $F(\Delta) - F(0) = \frac{\Upsilon}{2}\Delta^2$. It is positive below T_{BKT}, where the spins resist a twist as a superfluid resists a phase gradient (an analogy), and zero above. In the thermodynamic limit it jumps at T_{BKT} from the universal value $2T_{\mathrm{BKT}}/\pi$ to zero [36]. Below T_{BKT} it fixes the exponent of the correlation function, $G(r) \propto r^{-\eta}$ with $\eta = T/(2\pi\Upsilon)$, which reaches $1/4$ at the jump. On a finite lattice the jump is rounded, and the crossing $T^*(L)$ of $\Upsilon_L(T)$ with the line $2T/\pi$ approaches T_{BKT} from above. From the BKT form of ξ, a crossover where $\xi \approx L$ lies at $T(L) = T_{\mathrm{BKT}} + b^2/(\ln L)^2$, a straight line in $1/(\ln L)^2$. The crossings $T^*(L) = 0.936$, 0.921, 0.917 for $L = 16$, 32, 48 extrapolate in this way to 0.897 ± 0.003 (Figure 46). A

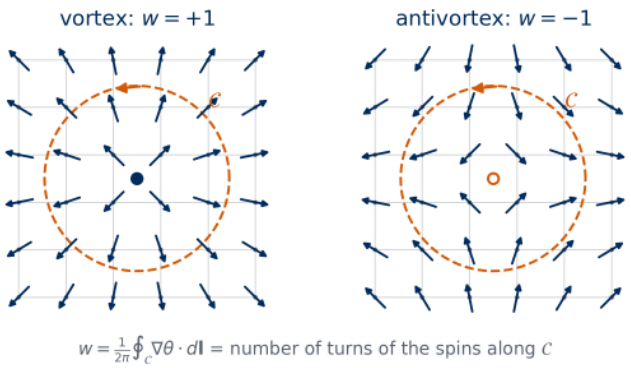


Figure 45: A vortex ($w = +1$, left) and an antivortex ($w = -1$, right) of planar spins (arrows) on a square lattice (grey). Along the closed path $\mathcal{C}$ (dashed) the spin direction turns once, counterclockwise for the vortex and clockwise for the antivortex.

detector that "finds the transition" can thus land on three different temperatures. These are the BKT transition located by the stiffness, the drop of the finite-lattice magnetisation, and the smooth rise of the vortex density and the energy near the specific-heat maximum. This maximum lies at $T = 1.117$, 1.067 and 1.060 for the three sizes.

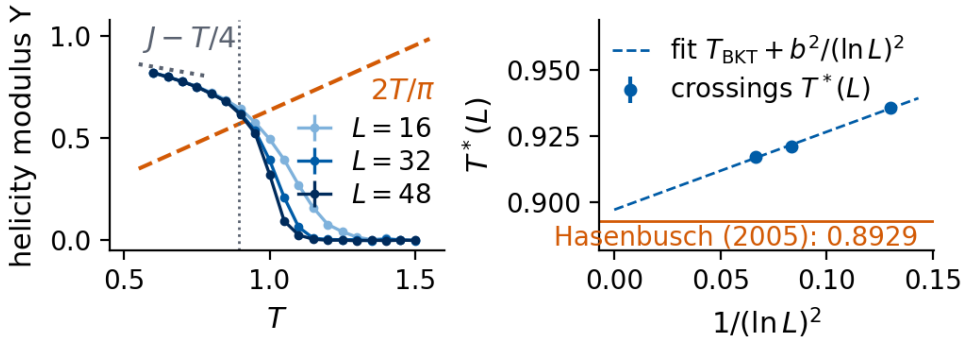


Figure 46: Spin stiffness of the XY model. Left, the helicity modulus Υ against T for $L = 16$, 32, 48, the line $2T/\pi$ (orange, dashed), the low-temperature approximation $J - T/4$ (grey, dotted) and T_{BKT} (vertical dotted line). Right, the crossings $T^*(L)$ of Υ_L with $2T/\pi$ against $1/(\ln L)^2$, with the straight-line fit $T_\infty + b^2/(\ln L)^2$ (dashed, fitted intercept T_∞) and the literature value of T_{BKT} (orange line).

10.4 Three representations of a snapshot

For $L = 16$, 32 and 48 the data contain 192 snapshots at each of 19 temperatures from 0.60 to 1.50. They come from 24 independent Monte Carlo chains that combine Metropolis updates with over-relaxation moves. The angle itself is a poor input for a neural network, since $\theta = 0.01$ and $\theta = 6.27$ describe almost the same spin. A snapshot can be presented in three ways, each an array with one or two numbers per site (Figure 47). The spin vectors $\mathbf{s}_r = (\cos\theta_r, \sin\theta_r)$ keep all information and rotate rigidly under a global rotation, so two snapshots that differ only by a rotation are physically identical and numerically different. The vortex map holds the winding number $w_r \in \{-1, 0, 1\}$ of every plaquette. Built from angle differences alone, it is invariant under global rotations and keeps the defects that drive the transition while discarding the spin waves. The bond-energy map holds the two bonds leaving every site, $\mathbf{b}_r = \big(\cos(\theta_{r+\hat{x}} - \theta_r), \cos(\theta_{r+\hat{y}} - \theta_r)\big)$, so that $H = -J\sum_r (b^x_r + b^y_r)$, and is the rotation-invariant local energy density. A supervised convolutional network fed with raw spins classifies the XY phases by learning the magnetisation, and vortex input changes what is learned [38]. We put the same test to the unsupervised methods.

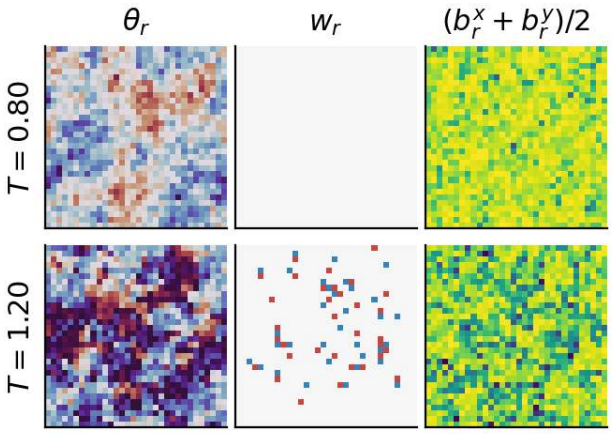


Figure 47: Three representations of one 32×32 snapshot at $T = 0.80$ (top) and at $T = 1.20$ (bottom). Left, the angle θ_r in a cyclic colour scale. Middle, the vortex map w_r, red for $w_r = +1$, blue for $w_r = -1$ and light grey for zero. Right, the mean bond energy $(b_r^x + b_r^y)/2$ of every site, yellow for values close to 1.

10.5 PCA and the finite-lattice magnetisation

PCA receives all 19×192 snapshots of one lattice size in one representation, without temperatures. Before the analysis every snapshot is rotated by its own random angle. The rotation uses the symmetry of H and removes a spurious mean direction left by chains that turn slowly at low temperature. For $L = 32$ the three spectra are very different (Figure 48). For the spin vectors two directions carry 13.38% and 13.16% of the variance, far above the next one, 0.95%. They span the plane of the two uniform patterns, the uniform cos part and the uniform sin part, whose projections on this plane have the length 0.9995. The projection on these directions is $\sqrt{N}(m_x, m_y)$ up to a rotation, and the equal variances express the rotation symmetry. For the bond map one direction dominates, 9.28% against 0.19%, and it is the uniform pattern, overlap 0.9986, whose projection is the energy of the snapshot. For the vortex map the spectrum is flat, every direction carrying about 0.34%. The uniform combination, the total vorticity, is zero on the torus. The binding of vortices in pairs is a property of correlations between defects, which no linear pattern expresses.

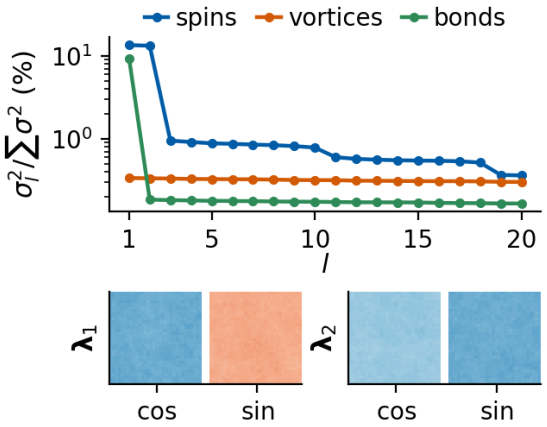


Figure 48: PCA of the XY snapshots for $L = 32$. Top, the fraction of the variance $\sigma_l^2 / \sum \sigma^2$ in per cent against the index l, on a logarithmic scale, for the spin vectors, the vortex maps and the bond-energy maps. Bottom, the first two principal directions of the spin vectors, each shown as its cos part and its sin part on the lattice, with one colour per sign.

The projection on the leading plane gives a two-component coordinate $\mathbf{w} = (w^{(1)}, w^{(2)})$ per snapshot, with the superscript in parentheses distinguishing it from the winding number w_r. Its length $\langle|\mathbf{w}|\rangle/\sqrt{N}$ agrees with $\langle|\mathbf{m}|\rangle$ of the same snapshots to within a few thousandths at every temperature. Large at low T and falling near $T \approx 1$ (Figure 49), it looks like an order parameter.

The temperature where it falls fastest, $T_{\mathrm{PCA}} = 1.129 \pm 0.015$, 1.062 ± 0.006 and 1.033 ± 0.006 for $L = 16$, 32, 48 (errors from resampling the independent chains), lies above T_{BKT} and drifts down as L grows. Its low-temperature plateau, however, decreases with L, so it fails as an order parameter of the XY model. A fit $\langle|\mathbf{w}|\rangle \propto L^{-x_{\mathrm{eff}}}$ to the three sizes gives an effective exponent x_{eff}. Up to T_{BKT} it follows the prediction $\eta/2$ computed from the measured stiffness, 0.064 against 0.058 at $T = 0.60$ and 0.128 against 0.117 at $T = 0.90$. A positive exponent means $\langle|\mathbf{w}|\rangle \to 0$ for $L \to \infty$, as the Mermin–Wagner theorem requires, although three sizes cannot prove it. PCA locates the temperature at which the correlation length becomes comparable with the lattice, a finite-size crossover that approaches T_{BKT} only slowly. The result agrees with the earlier finding that PCA of the XY model finds the magnetisation and misses the vorticity [39].

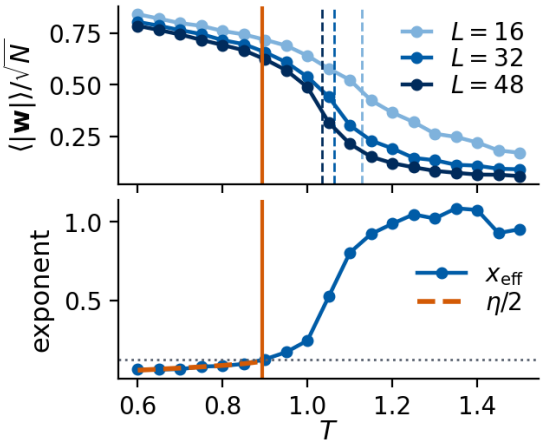


Figure 49: PCA coordinate of the XY spin vectors. Top, $\langle|\mathbf{w}|\rangle/\sqrt{N}$ against T for $L = 16$, 32, 48, with the steepest falls $T_{\mathrm{PCA}}(L)$ (dashed) and T_{BKT} (orange). Bottom, the effective exponent x_{eff} of $\langle|\mathbf{w}|\rangle \propto L^{-x_{\mathrm{eff}}}$ and the prediction $\eta/2 = T/(4\pi\Upsilon)$ from the stiffness of $L = 48$ (orange, dashed) against T. The dotted horizontal line marks 1/8, the value of $\eta/2$ at the jump.

10.6 Dependence of the autoencoder on the input

The autoencoder for the XY snapshots is convolutional with periodic padding. Two convolutions with stride 2 and 16 and 32 filters lead to a dense latent layer of 4 numbers. The decoder mirrors them with transposed convolutions and has a linear output, since the targets differ between representations. The same network, with 28 294 parameters for $L = 32$ and two channels, is used for every representation. Every training batch is rotated by fresh random angles, so that the network has no preferred direction to learn. The 24 chains are split, 12 for training and 12 for evaluation, and the training region is the low-temperature end $T \in \{0.60, 0.65, 0.70\}$, well below T_{BKT}. The estimate $T_{\mathrm{AE}}(L)$ is the position of the steepest rise of $\bar{\varepsilon}(T)$, the error of equation (8.5) averaged also over the channels.

On the spin vectors the error rises from its training level to about 1/2. It follows the error of the reconstruction that uses only the magnetisation of the snapshot, $\tilde{\mathbf{s}}_r = \mathbf{m}$ for every site,

$$\varepsilon_{\mathbf{m}} = \frac{1}{2N} \sum_r |\mathbf{s}_r - \mathbf{m}|^2 = \frac{1 - |\mathbf{m}|^2}{2}, \tag{10.4}$$

since $\frac{1}{N}\sum_r |\mathbf{s}_r|^2 = 1$ and $\frac{1}{N}\sum_r \mathbf{s}_r = \mathbf{m}$. The two curves correlate at 0.9999 or higher for all sizes (Figure 50). With four latent numbers and random rotations, the most useful information the network can store is the direction and length of $\mathbf{m}$. Its anomaly score inherits the behaviour of the PCA coordinate, $T_{\mathrm{AE}} = 1.130$, 1.038, 1.028 for the three sizes. Trained at the high-temperature end, $T \in \{1.40, 1.45, 1.50\}$, the network behaves like the paramagnetic regimes of the Ising and J_1–J_2 models. Learning little more than the mean, it reconstructs the magnetised low-temperature snapshots better than its own training data. The difference between the two ends shrinks with the size, and the relative range of $\bar{\varepsilon}$ falls from 0.32 for $L = 16$ to 0.14 for $L = 48$.

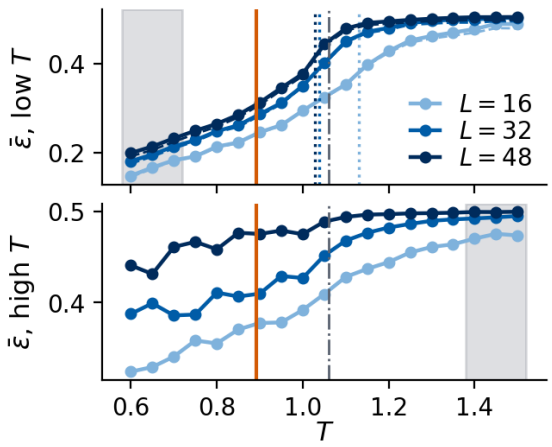


Figure 50: Autoencoder on the XY spin vectors. Mean reconstruction error $\bar{\varepsilon}$ against T for $L = 16$, 32, 48, trained at low T (top) and at high T (bottom), training temperatures shaded. Dashed (top), the magnetisation baseline $\langle(1 - |\mathbf{m}|^2)/2\rangle$. Dotted, the steepest rises $T_{\mathrm{AE}}(L)$. Orange, T_{BKT}. Dash-dotted, the specific-heat maximum for $L = 48$.

On the vortex map the network trained at low temperature has seen almost only empty maps, learns to output nearly zero, and its error approximates the vortex density, $\frac{1}{N}\sum_r w_r^2$. The vortex density rises smoothly and nearly independently of L. The estimates $T_{\mathrm{AE}} = 1.199$, 1.171, 1.170, with errors of 0.04 to 0.06, agree within two errors with the steepest rise of the vortex density itself, 1.154, 1.120, 1.104. These temperatures lie far above T_{BKT} and above the specific-heat maximum (Figure 51). With the finite-size magnetisation removed by the rotation-invariant input, the detector counts defects whether or not they are bound in pairs. On the bond map the error follows the spatial fluctuations of the local energy. Its steepest rise, 1.146, 1.047, 1.041, lies within its errors on the steepest rise of the energy, which is the specific-heat maximum.

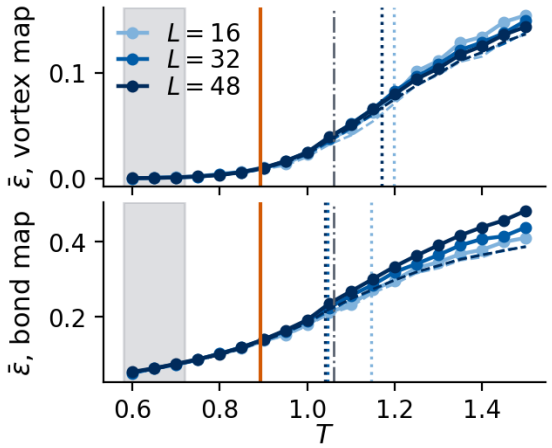


Figure 51: Autoencoder trained at low T (shaded) on the vortex maps (top) and on the bond-energy maps (bottom). Mean reconstruction error $\bar{\varepsilon}$ against T for $L = 16$, 32, 48. The dashed baselines are the vortex density for the vortex map and the spatial variance of the bond energies for the bond map. Dotted, the steepest rises $T_{\mathrm{AE}}(L)$. Orange, T_{BKT}. Dash-dotted, the specific-heat maximum for $L = 48$.

10.7 Network size and finite-size extrapolation

A network that is too small or trained too briefly could miss everything beyond the magnetisation. For $L = 32$ we vary one property at a time, the latent size (2, 8, 16), the number of filters (doubled or quadrupled) and the training time (tripled). The final loss changes, from 0.209 with 2 latent numbers to 0.173 with 16, while every estimate of the spin autoencoder stays between 1.036 and 1.039. The

largest shift from the baseline is 0.14 times the combined error for the spin autoencoder and 0.10 times for the vortex autoencoder. A larger network reconstructs slightly better and measures the same thing.

The unsupervised temperatures could also approach T_{BKT} for $L \to \infty$, since even the stiffness crossings lie above T_{BKT} on every finite lattice and reach it only after extrapolation. Fitting $T(L) = T_\infty + b^2/(\ln L)^2$ to the PCA temperatures gives $T_\infty = 0.932 \pm 0.020$, and fitting it to the spin autoencoder temperatures gives 0.952 ± 0.036 (Figure 52). With T_∞ fixed at T_{BKT} the fits give $\chi^2 = 4.22$ and 4.15 for two degrees of freedom, with p-values 0.12 and 0.13. Here χ^2 is the sum of squared deviations in units of the errors. The p-value is the probability of a χ^2 at least this large if the form is correct. The magnetisation crossover is therefore consistent with a drift towards T_{BKT}. This drift is expected, since the finite-lattice magnetisation drops where ξ reaches L and ξ diverges at T_{BKT}. The same extrapolation of the specific-heat maximum gives 0.995, so the fitted form does not force every set of three numbers onto T_{BKT}. Three sizes and two parameters leave one degree of freedom, so the fits are equally compatible with a limit near 0.93 to 0.95. The form $b^2/(\ln L)^2$ itself comes from knowing beforehand that the transition is of the BKT type.

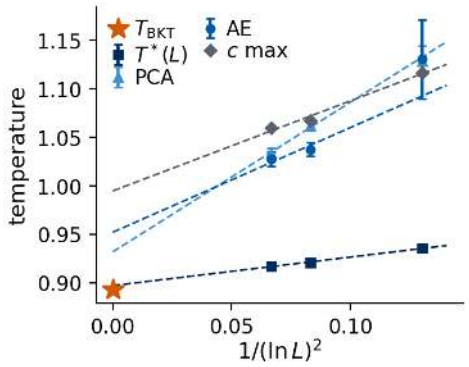


Figure 52: Temperatures against $1/(\ln L)^2$ with straight-line fits (dashed), for the stiffness crossings $T^*(L)$, PCA, the autoencoder on spins (AE) and the specific-heat maximum. The star marks T_{BKT} at $L \to \infty$.

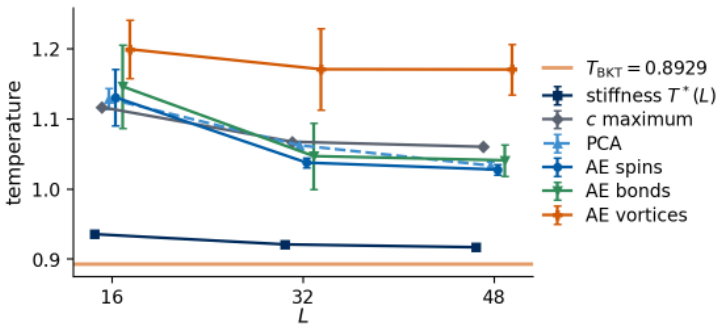


Figure 53: All temperatures of the XY analysis against L, with bootstrap errors, and T_{BKT} (horizontal line). They are the stiffness crossings $T^*(L)$, the specific-heat maximum, the steepest fall of the PCA coordinate and the steepest rises of the three autoencoder errors. Points are shifted slightly in L for visibility.

For $L = 48$ all unsupervised temperatures of Figure 53 lie between 1.028 and 1.170, 0.14 to 0.28 above T_{BKT} and well above the stiffness crossing $T^*(48) = 0.917$. Each method finds the quantity that its input makes easy to see, the magnetisation of the finite lattice, the vortex density or the energy. All three change smoothly through T_{BKT}. The defining property of the low-temperature phase, the binding of vortices into pairs and the resulting stiffness, is a response to a twist. No single map of a snapshot displays it as a pattern. Only the physics identifies the temperatures found as a finite-size crossover and two smooth features, and T_{BKT} has to be obtained from the stiffness.

11 Order without a local order parameter

In the XY model each detector found what changes most in its input. In the Ising lattice gauge theory a theorem forbids a local order parameter, and in a spin glass the order is invisible in a single snapshot.

11.1 The Ising lattice gauge theory

In the Ising lattice gauge theory the variables $\sigma_\mu(x) = \pm 1$ sit on the links of a periodic lattice of $N = L^d$ sites in d dimensions. Each variable belongs to one link, the link from site x to its neighbour $x + \hat{e}_\mu$ in direction μ (Figure 54). The product of the four links around an elementary square p, a

plaquette, is $U_p \coloneqq \sigma_\mu(x)\,\sigma_\nu(x+\hat{e}_\mu)\,\sigma_\mu(x+\hat{e}_\nu)\,\sigma_\nu(x) = \pm 1$. The energy is

$$H = -J \sum_p U_p, \tag{11.1}$$

where the coupling is written as $\beta = J/T$, so that a large β means a low temperature [40]. The PCA variances σ_l^2 are unrelated to the links $\sigma_\mu(x)$. The model has a local symmetry. Choosing $\gamma(x) = \pm 1$ at every site and replacing $\sigma_\mu(x) \to \gamma(x)\sigma_\mu(x)\gamma(x+\hat{e}_\mu)$ leaves every plaquette unchanged, since each γ appears squared. Flipping the sign at one site flips the links that touch it, and the configurations related by such gauge transformations form an orbit of 2^{N-1} physically equivalent configurations. Elitzur's theorem states that a local symmetry cannot be broken spontaneously [41], so $\langle \sigma_\mu(x) \rangle = 0$ on every link at every temperature. The average of any product of links that do not form closed paths vanishes as well. The covariance matrix of the raw links is the identity matrix in every phase.

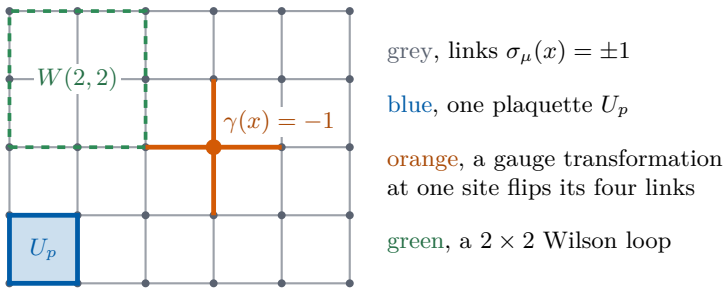


Figure 54: The Ising lattice gauge theory in two dimensions, sketch. Variables live on the links of the lattice (grey). A plaquette variable U_p is the product of the four links around a square (blue). Changing the sign $\gamma(x)$ at one site flips the four links that touch it (orange) and leaves every plaquette unchanged. A Wilson loop is the product of the links along the boundary of a rectangle (green, dashed).

The phases differ in how Wilson loops decay. The Wilson loop $W(R, S)$ of an $R \times S$ rectangle is the thermal average of the product of the links along its boundary [42]. It equals the product of the plaquettes it encloses, so $W(1, 1) = \langle U_p \rangle$. In the confined phase it decays with the area, $W \sim e^{-\sigma_s RS}$ with a string tension $\sigma_s > 0$, and in the deconfined phase with the perimeter, $W \sim e^{-2\kappa(R+S)}$ with a constant $\kappa > 0$. In two dimensions the plaquettes are independent and $\langle U_p \rangle = \tanh \beta$. The specific heat $\beta^2 / \cosh^2 \beta$ then has a smooth maximum, and the model has no transition. In three dimensions Wegner's duality to the three-dimensional Ising model locates a continuous confinement–deconfinement transition at $\beta_c = 0.7614$ [40], with the area law for $\beta < \beta_c$ and the perimeter law for $\beta > \beta_c$. The data are heat-bath Monte Carlo configurations for $L = 16$ and 32 in two dimensions and for $L = 6$, 8 and 10 in three dimensions. The three-dimensional data cover 29 couplings between 0.3 and 1.2, refined to steps of 0.01 around β_c. Every stored configuration was subjected to an independent random gauge transformation, which changes nothing physical. The maximum of the specific heat per plaquette lies at $\beta = 0.7145$, 0.7326 and 0.7406 for $L = 6$, 8 and 10, below β_c and moving towards it.

On the raw links the unsupervised methods find nothing (Figure 55). For $L = 8$ in three dimensions we pool 3712 configurations over all couplings, each a vector of $3L^3 = 1536$ links. PCA gives the largest variance 2.69 and the median 0.86. A matrix of random ± 1 entries of the same shape gives the largest variance 2.67 and the same median, so the two are indistinguishable. The first coordinate has the same distribution at every coupling. A dense autoencoder like that of the Ising model, trained on the most ordered configurations, $\beta \geq 1.0$, reconstructs nothing, and its error stays between 1.0075 and 1.0093 at every coupling. In a random gauge a configuration is a uniformly random element of its gauge orbit. Even a fully ordered state with all $U_p = +1$ appears as $N - 1$ random bits spread over the links, which no latent vector of small dimension can rebuild.

Replacing the links by the gauge-invariant plaquettes removes the redundancy. The plaquette spectrum has one eigenvalue 66.9 times larger than the second, carrying 15.5% of the variance. Its direction is the uniform vector (overlap 0.9984). The projection on it correlates with the plaquette average $\bar{U}$ of the configuration with the coefficient 0.99999, so PCA has found the energy. We train an autoencoder on plaquettes either deep in the deconfined phase ($\beta \geq 1.0$) or deep in the confined phase

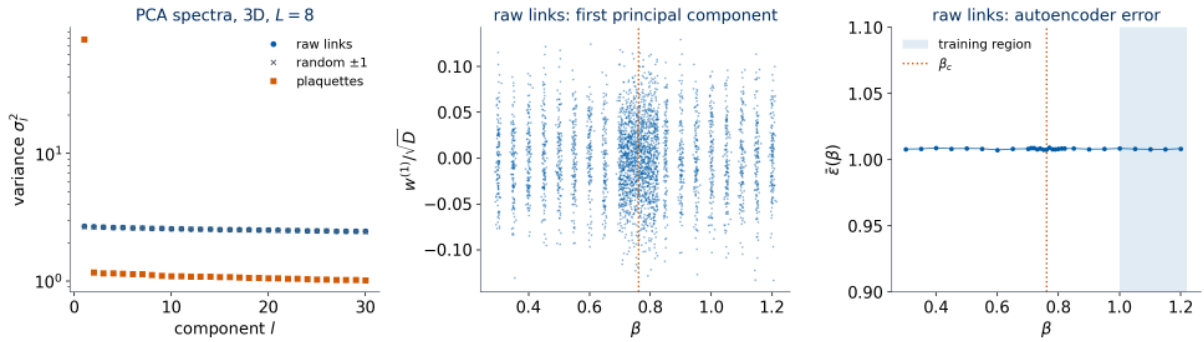


Figure 55: Raw links of the three-dimensional gauge theory, $L = 8$. Left, the leading variances σ_l^2 against the index l for raw links, for random ± 1 entries and for the plaquettes of the same configurations. Middle, the first principal coordinate $w^{(1)}/\sqrt{D}$ of every raw-link configuration ($D = 1536$) against its coupling β. Right, the mean reconstruction error $\bar{\varepsilon}(\beta)$ of the autoencoder trained on raw links with $\beta \geq 1.0$ (shaded). Dotted, β_c.

($\beta \leq 0.45$). Its error falls steeply between $\beta \approx 0.6$ and 0.8, with the steepest change at $\beta^* = 0.715$, 0.735 and 0.745 for $L = 6$, 8 and 10 and both training regions. These values lie in the grid interval in which $\langle U_p \rangle$ rises fastest, within 0.005 of the specific-heat peak, below β_c and moving towards it. Plotted against $\langle U_p \rangle$ the errors of all sizes collapse onto one curve per training region (Figure 56). For the deconfined training this curve is close to $2(1 - \langle U_p \rangle)$, the error of a network that outputs $+1$ for every plaquette, so the detector measures the energy.

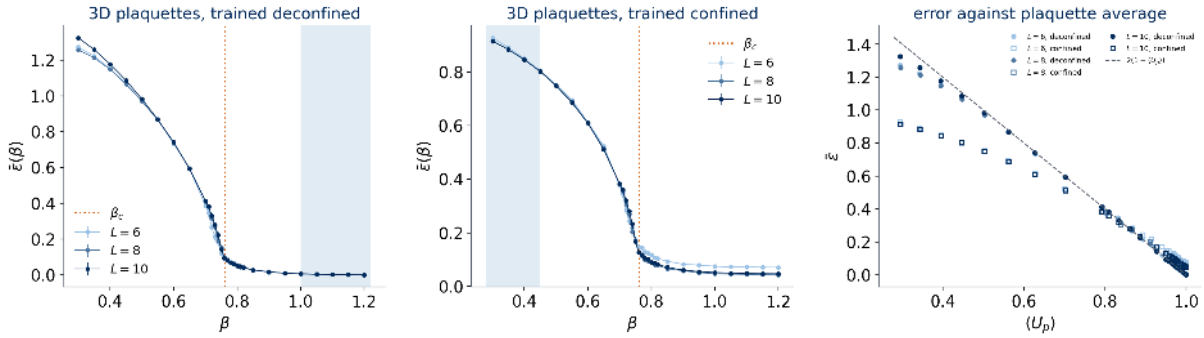


Figure 56: Autoencoder on the plaquettes of the three-dimensional gauge theory. Left and middle, the mean reconstruction error $\bar{\varepsilon}(\beta)$ for $L = 6$, 8, 10, trained on the deconfined phase $\beta \geq 1.0$ and on the confined phase $\beta \leq 0.45$ (shaded). Dotted, β_c. Right, the same errors against the plaquette average $\langle U_p \rangle$, with the line $2(1 - \langle U_p \rangle)$ (dashed).

A steep rise of the energy is a real signature of the transition, although an energy detector would find the same signature at a smooth crossover. The two-dimensional model, which has no transition, serves as the control (Figure 57). Its error has a broad region of steepest descent between $\beta \approx 0.4$ and 0.9 that stays in place and keeps its width from $L = 16$ to $L = 32$. In three dimensions the steepest change moves with L like the specific-heat peak. Its maximal slope shows no systematic growth with L in these data, 6.27, 6.46, 6.10 for the deconfined training region, so the sharpening remains unresolved. Only the size dependence separates the transition from the crossover, and an energy detector is at most as sharp as the energy.

At β_c the area law, a relation between loops of different sizes, is lost. PCA of the standardised loop averages $(\bar{W}(1,1), \bar{W}(2,2), \bar{W}(3,3), \bar{W}(4,4))$ of each configuration finds the energy a second time. The component with 95.5% of the variance weights the loops nearly equally and correlates with $\bar{U}$ at 0.95. The physics prescribes the combination to build. If $-\ln W(R,R) \propto R^{a_W}$, then $a_W = 2$ for an area law and $a_W = 1$ for a perimeter law, and the two smallest square loops give the loop-shape exponent $a_W := \log_2 [\ln W(2,2) / \ln W(1,1)]$. The Creutz ratio [43]

$$\chi_C(2,2) := -\ln \frac{W(2,2)\, W(1,1)}{W(2,1)^2} \tag{11.2}$$

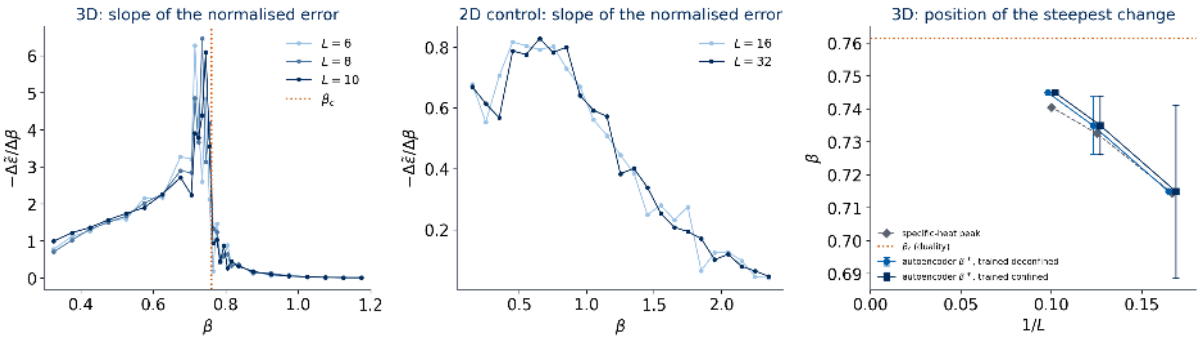


Figure 57: Steepness of the plaquette autoencoder error. Left, the slope $-\Delta\tilde{\varepsilon}/\Delta\beta$ of the error against β, in three dimensions for $L = 6$, 8, 10, with β_c dotted. The error is normalised to run from 1 at the smallest to 0 at the largest coupling. Middle, the same for the two-dimensional control, $L = 16$ and 32. Right, the position β^* of the steepest change in three dimensions for both training regions and the specific-heat peak, against $1/L$, with β_c dotted. The error bars of the positions are jackknife errors over the Monte Carlo chains.

cancels the perimeter and constant terms of $\ln W$ and estimates the string tension. In two dimensions a_W stays between 1.97 and 2.01 at all couplings, the area law everywhere. In three dimensions it falls from 2 towards 1 and passes 1.5 at $\beta = 0.742$, 0.751 and 0.752 for the three sizes, close to β_c. The Creutz ratio for $L = 10$ falls from 0.415 at $\beta = 0.60$ to 0.0125 at 0.76 and 0.0004 at 0.90 (Figure 58). These combinations encode what we know about confinement, and no unsupervised method discovered them. The same was found in a supervised setting [44]. A fully connected network on raw gauge configurations stayed at chance level. A convolutional network with 2×2 filters, which cover the four links of a plaquette, succeeded by computing the plaquette constraint, which is the energy.

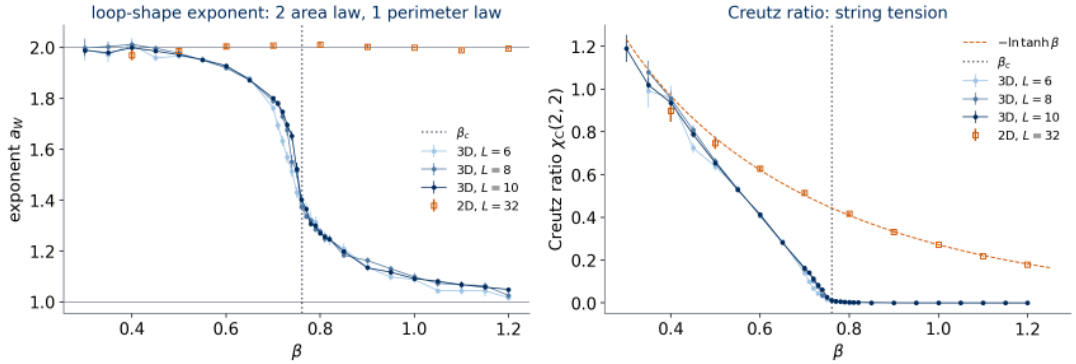


Figure 58: Wilson-loop combinations of the gauge theory against β. Left, the loop-shape exponent a_W in three dimensions for $L = 6$, 8, 10 and in two dimensions for $L = 32$. It is 2 for an area law and 1 for a perimeter law (horizontal lines). Right, the Creutz ratio $\chi_C(2, 2)$ for the same data and the exact two-dimensional value $-\ln\tanh\beta$ (dashed). Dotted, β_c.

11.2 The Sherrington–Kirkpatrick spin glass

In the Sherrington–Kirkpatrick (SK) model [45] every pair of N Ising spins is coupled by a random coupling drawn once and then kept fixed, a situation called quenched disorder,

$$H = -\sum_{i<j} J_{ij} s_i s_j, \qquad J_{ij} \sim \mathcal{N}(0, J^2/N), \tag{11.3}$$

with $J = 1$. The scaling $1/N$ of the variance keeps the local field on a spin of order J. About half of the couplings are negative. A triangle of spins is frustrated when the product of its three couplings is negative, which happens for half of the triangles. One draw of all couplings is called a disorder sample. Square brackets $[\,\cdot\,]$ denote the average over disorder samples and angular brackets the thermal average in one sample. Below $T_c = J$ the spins freeze in random directions, the magnetisation stays zero, and

a snapshot of the frozen phase looks like noise. The order shows only when two replicas, independent copies with the same couplings, are compared through their overlap

$$q \coloneqq \frac{1}{N}\sum_{i=1}^{N} s_i^a s_i^b. \tag{11.4}$$

In one valley of the free energy q^2 equals the square of the Edwards–Anderson order parameter $q_{\mathrm{EA}} \coloneqq \frac{1}{N}\sum_i [\langle s_i \rangle^2]$ [46]. Above T_c the overlap distribution $P(q)$ is a single peak at $q = 0$ that narrows as N grows. Below T_c it has two peaks and a weight between them that stays finite as N grows, the signature of many valleys [47] (Figure 59). The transition is continuous, of mean-field type, and signalled by the divergence of the spin-glass susceptibility $\chi_{\mathrm{SG}} \coloneqq N[\langle q^2 \rangle] = 1/(1-\beta^2 J^2)$ as $T \to J$ from above. We locate it with the Binder cumulant of the overlap,

$$g_{\mathrm{B}} \coloneqq \frac{1}{2}\left(3 - \frac{[\langle q^4 \rangle]}{[\langle q^2 \rangle]^2}\right), \tag{11.5}$$

which is 0 for a Gaussian $P(q)$ and 1 for two narrow peaks, and whose curves for different N cross at T_c. The data are snapshots of two replicas for $N = 32$, 64 and 128, with 256, 160 and 96 disorder samples and 25 temperatures from 0.40 to 1.60. They come from parallel-tempering Monte Carlo, which simulates copies at a ladder of temperatures that exchange configurations. A configuration trapped in a valley at low temperature can thus escape through high temperatures. The Binder curves cross at 0.870 for $N = 32$ and 64, at 0.920 for 64 and 128 and at 0.894 for 32 and 128. The 68% bootstrap intervals over disorder samples are $[0.765, 0.933]$, $[0.816, 0.999]$ and $[0.848, 0.938]$. The crossings lie somewhat below $T_c = J$, and the intervals are broad, because the curves meet at a small angle and finite-size corrections shift the crossings for sizes up to $N = 128$.

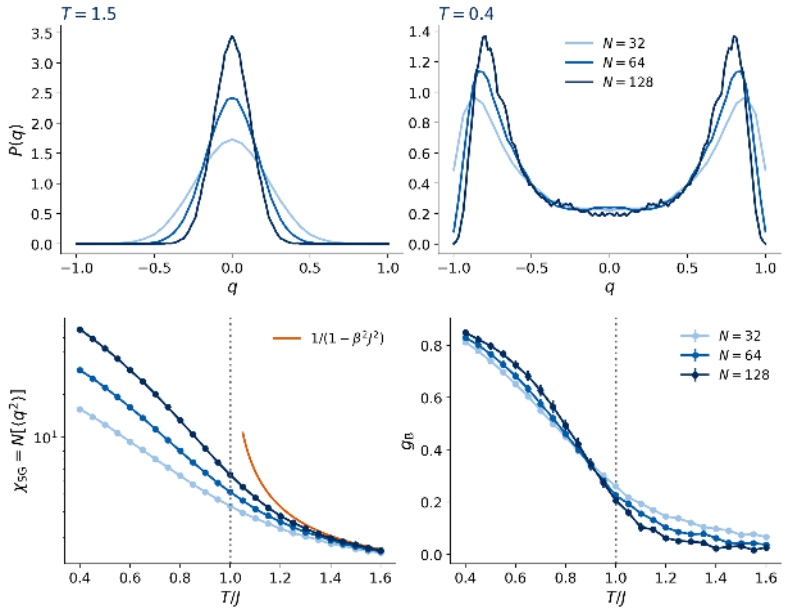


Figure 59: Physics of the SK spin glass. Top, the overlap distribution $P(q)$ as a density at $T = 1.5$ and $T = 0.4$ for $N = 32$, 64, 128. Bottom left, the spin-glass susceptibility $\chi_{\mathrm{SG}} = N[\langle q^2 \rangle]$ against T/J on a logarithmic axis, with $1/(1-\beta^2 J^2)$ (orange). Bottom right, the Binder cumulant g_{B} against T/J. Dotted, $T_c = J$.

One snapshot carries no signal of this order. The transformation $s_i \to \gamma_i s_i$, $J_{ij} \to \gamma_i \gamma_j J_{ij}$ with $\gamma_i = \pm 1$ leaves both the energy and the Gaussian distribution of the couplings unchanged. Pooled over disorder samples, a snapshot is therefore a uniformly random string of signs at every temperature. With one snapshot per sample the largest PCA variance for $N = 128$ stays between 4.23 and 4.72. These values lie within two standard deviations of the value 4.46 ± 0.15 for random strings of the same shape (Figure 60). With all 64 snapshots of each sample it grows from 1.37 at $T = 1.6$ to 3.35 at $T = 0.4$, against 1.30 for random strings. This growth comes from repeated snapshots of the

same sample frozen in related valleys, so it measures indirectly the overlap between snapshots of one sample. An autoencoder trained on frozen snapshots reconstructs its own training samples well at low temperature, while on new samples its error is independent of temperature. The frozen patterns it has stored belong to particular samples and fail to transfer to a new sample.

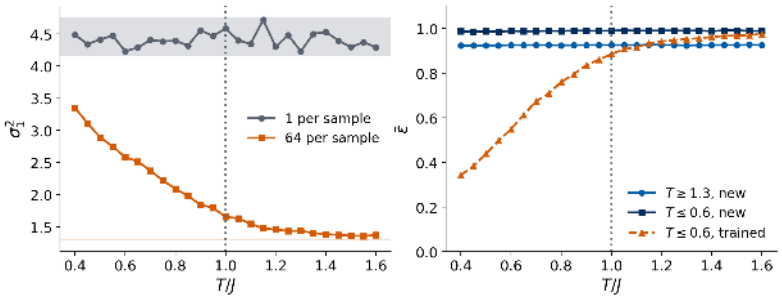


Figure 60: Single SK snapshots, $N = 128$. Left, the largest PCA variance σ_1^2 against T/J for one snapshot per disorder sample (grey) and for 64 snapshots per sample (orange). The shaded band (grey) and the light line (orange) give the values for random strings. Right, the mean reconstruction error $\bar{\varepsilon}$ of autoencoders trained on $T \geq 1.3$ and on $T \leq 0.6$, evaluated on new disorder samples (solid) and, for $T \leq 0.6$, on the training samples (dashed). Dotted, $T_c = J$.

The right input is the overlap field of two replicas, $q_i := s_i^a s_i^b$, which is invariant under the transformation above. PCA of the overlap fields pooled over samples and temperatures gives a first direction equal to the uniform vector, so that $w^{(1)} = \sqrt{N}\, q$. For $N = 128$ it carries 11.0% of the variance, against below 1% for each of the others (Figure 61). PCA has found the overlap, as it found the magnetisation of the Ising model. The Binder cumulant of $w^{(1)}$ crosses at 0.869, 0.920 and 0.903, within the intervals of the reference. An autoencoder with a single latent number, trained on the frozen phase $T \leq 0.6$, learns the overlap. A network that encodes q and outputs q at every site has the error $\frac{1}{N}\sum_i (q_i - q)^2 = 1 - q^2$, and the mean error follows $1 - [\langle q^2 \rangle]$ to within 0.0043 for $N = 128$ (Figure 62). The Edwards–Anderson parameter grows linearly just below T_c, $q_{\mathrm{EA}} \approx 1 - T/J$, so q^2 starts quadratically and the error has no edge at T_c. Its steepest change lies at $T = 0.475$, 0.425, 0.475 for the three sizes, far below T_c and without a drift towards it.

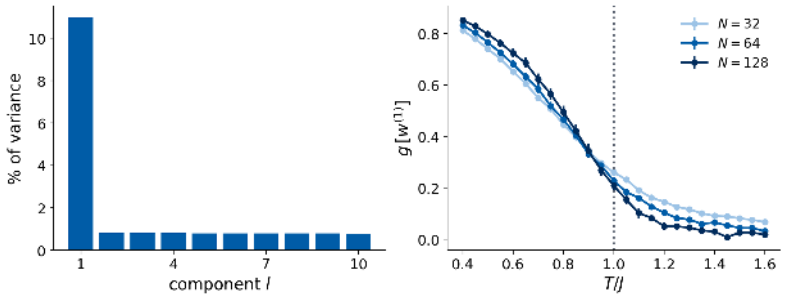


Figure 61: PCA of the SK overlap fields. Left, the percentage of the variance carried by the first ten principal directions, $N = 128$. Right, the Binder cumulant of the first principal coordinate, $g_{\mathrm{B}}[w^{(1)}]$, against T/J for $N = 32$, 64, 128. Dotted, $T_c = J$.

A detector can also be given the histogram of $|q|$ over all replica pairs of one sample and temperature and be trained on the paramagnet, $T \geq 1.3$. Its anomaly score is the error in units of its spread on the training data. It reaches 3 at $T = 1.007$, 1.094 and 1.167 for $N = 32$, 64 and 128, above T_c and increasing with N. Above T_c the width of $P(q)$ grows continuously as T approaches J, as χ_{SG} does. The detector raises the alarm as soon as this growth exceeds the sample-to-sample noise of its training histograms, a noise that decreases with N. An anomaly onset depends on the threshold, the training window and the resolution of the data. A mean-field transition approached through a smooth growth of fluctuations gives no abrupt edge.

The Mattis model [48], with couplings $J_{ij} = \frac{J}{N}\zeta_i\zeta_j$ of random fixed signs $\zeta_i = \pm 1$, serves as the control. Its couplings look as random as those of the SK model. The transformation $s'_i = \zeta_i s_i$ turns

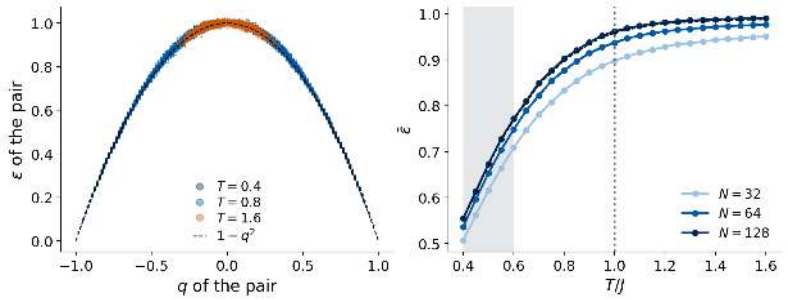


Figure 62: Autoencoder with one latent number on the SK overlap fields, trained on $T \leq 0.6$ (shaded) and evaluated on new disorder samples. Left, the reconstruction error ε of individual replica pairs against their overlap q at three temperatures, with the parabola $1 - q^2$ (dashed). Right, the mean error $\bar{\varepsilon}$ against T/J for $N = 32$, 64, 128, with $1 - [\langle q^2 \rangle]$ for $N = 128$ (dashed). Dotted, $T_c = J$.

it into a ferromagnet with every pair coupled, so it has random signs and no frustration. With the same detectors its Binder crossings 0.927, 0.947, 0.940 have intervals more than ten times narrower, for instance $[0.938, 0.944]$ for $N = 32$ and 128. The autoencoder error falls to 0.059 at $T = 0.4$ for $N = 128$, against 0.555 for the SK model. The histogram onsets lie above T_c and increase with N as for the SK model (Figure 63). The broad intervals and the large low-temperature error of the SK model, absent in the Mattis model, are thus attributed to frustration.

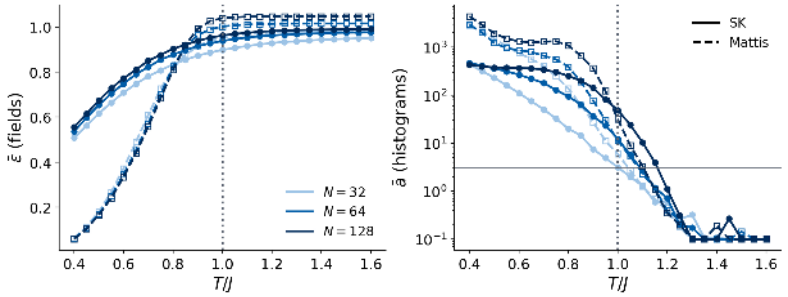


Figure 63: SK model (solid, filled circles) and Mattis model (dashed, open squares) for $N = 32$, 64, 128. Left, the mean reconstruction error $\bar{\varepsilon}$ of the overlap-field autoencoder against T/J. Right, the mean anomaly score $\bar{a}$ of the histogram detector trained on $T \geq 1.3$, with the level 3 (horizontal line). On the logarithmic axis values below 0.1 are drawn at 0.1. Dotted, $T_c = J$.

12 The common protocol and its limits

Table 3 collects what PCA and the autoencoders found in the five models and what the physics had to add.

Both methods find the structure that changes most in the input they are given. PCA finds the order parameter when it is linear in that input. The magnetisation and the stripe order are linear in the spins, and the overlap is linear in the overlap field. Its projection can then be analysed with finite-size scaling. The autoencoder finds where the snapshots stop resembling a reference region. In the Ising and J_1–J_2 models this marks the phase boundaries at a small fraction of the cost of a dense scan. The outcome depends on choices made outside the method. The magnetisation missed the stripe phase, the vortex map gave the vortex density and the raw gauge links gave nothing, so the input representation determines what can be seen. The training region determines the signal, and a structured, well-understood phase is the right reference. A temperature found on one lattice remains a finite-size crossover until its size dependence has been examined. In the gauge theory only this dependence separated a transition from a smooth crossover. In each case physics supplied which order sets in, whether a transition is continuous or first order, and which quantity is the right input. An unsupervised method serves as a survey that locates where to measure more carefully. The less it is told, the less of the answer comes from us, and explaining the result remains our task.

Table 3: What PCA and the autoencoder found in the five models, compared with the known physics. Temperatures for the largest lattice of each data set.

model	order	what the methods found	what the physics added
2D Ising	magnetisation, linear in the spins	PCA gives a uniform $\boldsymbol{\lambda}_1$ and $w^{(1)} \propto m$, the autoencoder $T_{\mathrm{AE}} = 2.2641$ ($L = 50$), exact $T_c = 2.2692$	finite-size scaling, the type of transition
J_1–J_2	m and m_{str}, both linear in the spins	PCA gives both order parameters, the autoencoders the boundary of each training phase, a coarse scan at 0.5% of the cost	which order sets in, the type of transition near $g = 1/2$
2D XY	no local order parameter, vortex unbinding at $T_{\mathrm{BKT}} = 0.8929$	the finite-lattice magnetisation, the vortex density or the energy, 1.028 to 1.170 ($L = 48$)	the stiffness locates T_{BKT}
Ising gauge theory	none by Elitzur's theorem, area against perimeter law at $\beta_c = 0.7614$	nothing from raw links, the energy from plaquettes, $\beta^* = 0.745$ ($L = 10$)	gauge-invariant input, Wilson-loop combinations, the 2D control
SK spin glass	Edwards–Anderson overlap, invisible in one snapshot	nothing from single snapshots, from overlap fields PCA gives q and the autoencoder learns $1 - q^2$	replicas, finite-size scaling of g_{B}

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